

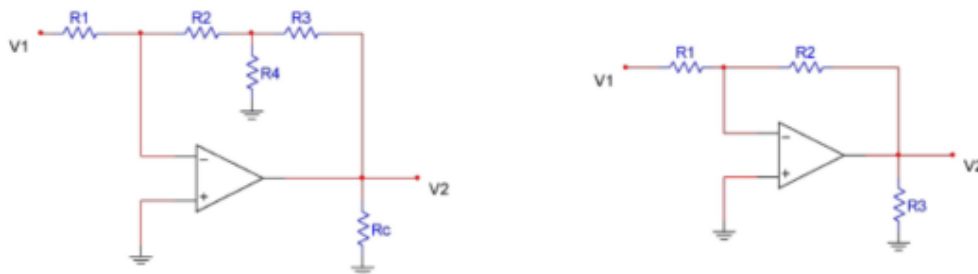
CIRCUIT THEORY II - R4053 - 2025**Professor: Navarro, Carlos****ATPs: Balbi Juan - Howlin
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Exercise #1

It is required to design inverting amplifiers with $Z_e = 10\text{ k}\Omega$ and $AV = 3000$ from the circuits in image 1. Simulate and analyze both advantages and disadvantages.



Circuit structures specified to implement the inverting amplifier.

Imagen 1

Left Circuit to):

Applying Kirchhoff's Current Law and using the impedance expressions of the elements, we see that:

$$OFF = \frac{V_2}{V_1} = - \frac{R_2}{R_1} * [1 + R_3 * \frac{(R_2+R_4)}{R_2*R_4}]$$

Then, in order for $AV = 3000$, we assume as a design criterion that, for example, the factor $\frac{R_2}{R_1} = 10$ and that the factor $[1 + R_3 * \frac{(R_2+R_4)}{R_2*R_4}] = 300$ where the input impedance is $Z_e = 10\text{ k}\Omega = R_1$. Then, from $\frac{R_2}{R_1} = 10$ we clear R_2 , so we get $R_2 = 10 * R_1 = 10 * 10\text{ k}\Omega = 100\text{ k}\Omega$. In turn, we impose that $R_3 = 100\text{ k}\Omega$ so that we are left with the other factor as $[1 + 100\text{ k}\Omega * \frac{(100\text{ k}\Omega+R_4)}{100\text{ k}\Omega*R_4}] = 300$ and solving for R_4 we get a value of $R_4 = 335.57\Omega$ which approximates $R_4 = 330\Omega$. Using commercial values, we then have:

• $R_1 = 10\text{ k}\Omega$, • $R_2 = 100\text{ k}\Omega$, • $R_3 = 100\text{ k}\Omega$, • $R_4 = 330\Omega$, all with a 5% tolerance.

Advantages:

- The input impedance Z_e can be controlled with resistor R_1 , affecting only a portion of the total gain. The gain then depends on two terms: $\frac{R_2}{R_1}$ and $[1 + R_3 * \frac{(R_2+R_4)}{R_2*R_4}]$ which gives us more flexibility to control the profit.
- Noise reduction since, depending on the resistance values, thermal noise can be reduced.
- Reducing the effects of the generator's output impedance relative to the circuit's input impedance in terms of power and energy. This is achieved by adjusting the circuit's input impedance to match the generator's output impedance, thus maximizing energy transfer.

Disadvantages:

- Multiple components on the PCB, which can increase manufacturing costs.
- Reduction of bandwidth due to the parasitic capacitive effects inherent to resistors.
- Greater sensitivity to tolerance errors in terms of percentage variability in resistance values. More resistors mean more potential variations in values due to tolerances specified by manufacturers.
- The input impedance is set by R1, so in some designs the chosen value of R1 may not be optimal.

Simulation:

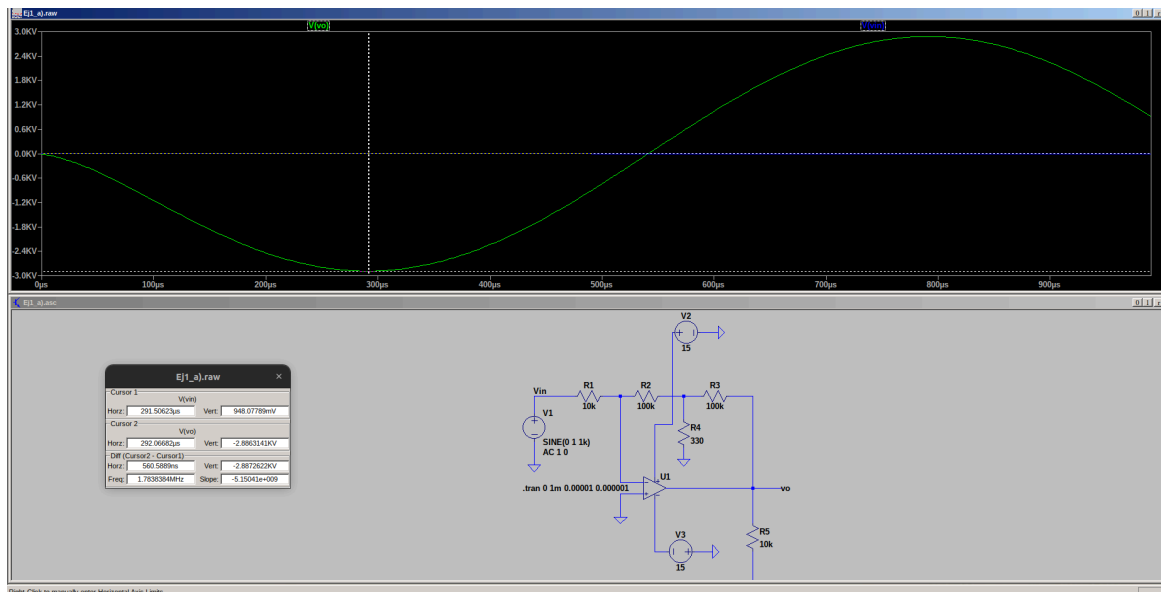


Imagen 2

Simulation of the voltage gain of the circuit on the left.

Dotted Line → Phase

Full Line → Module

By injecting an input sinusoidal signal, we see that approximately $3000 \text{ V} = v_o$ is reached with respect to $1 \text{ V} = V_1$, which gives an $AV = 3000$ at least at low frequencies less than or equal to 1 kHz . Then, a reduction in the gain factor becomes increasingly noticeable for input signals with frequencies greater than 1 kHz ; that is, as the frequency increases, the signal is amplified but with less gain.

We also see that the phase is inverted, a key characteristic of the inverting active filter. This phase inversion begins at 180° , suggesting a phase shift of π , that is, a phase inversion. This is true at least for low frequencies, below 1 kHz . For frequencies above 1 kHz , there is a phase shift, but it is less than 180° , which is not a strict phase inversion.

Right circuit b):

Applying Kirchhoff's Current Law and using the impedance expressions of the elements, we see that:

$$OFF = \frac{V_2}{V_1} = - \frac{R_2}{R_1}$$

Then, so that $AV = 3000$, we make the expression $\frac{R2}{R1} = 3000$ where the input impedance is $Z_e = 10 \text{ k}\Omega = R1$. Then, from $\frac{R2}{R1} = 10$ We clear $R2$, so we get $R2 = 3000 * R1 = 3000 * 10\text{k}\Omega = 30 \text{ M}\Omega$. There is no trade value of $30\text{M}\Omega$ But it is possible to form $R3$ with 3 series resistance values of $10\text{M}\Omega$ each one. Using commercial values we then have:

• $R1 = 10\text{k}\Omega$, • $R2 = 30 \text{ M}\Omega$

Advantages:

- Simplicity of design and circuit since there are only 2 components that define the gain, which implies a simpler, cheaper and easier to analyze PCB.
- Having fewer components means less variability in resistance values due to the tolerance percentages provided by the manufacturers.
- Greater bandwidth because by reducing the number of resistors in the feedback stage, the parasitic capacitive effects of the resistors can be minimized, thus improving the frequency response.

Disadvantages:

- The input impedance is set by $R1$, so in some designs the chosen value of $R1$ may not be optimal.
- Less stability because there is no compensation network such as a resistive divider or something to control stability.
- Greater susceptibility to noise from the excitation source, because if $R1$ is not chosen properly, then the source can introduce noise into the circuit.

Simulation:

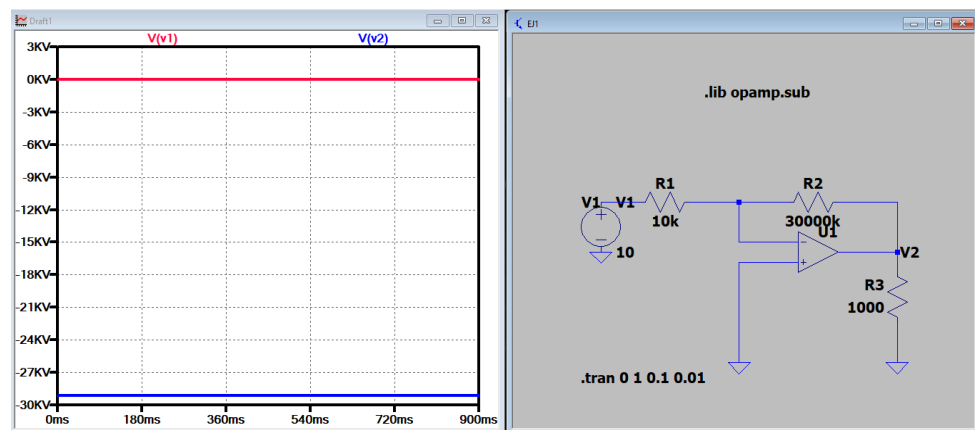


Imagen 3

Simulation of the voltage gain of the circuit on the right.

Dotted Line → Phase

Full Line → Module

Now, injecting a DC input, we see that approximately $3000 \text{ V} = v_o$ is reached with respect to $1 \text{ V} = V1$, resulting in an $AV = 3000$. We also observe a phase inversion, a key characteristic of the inverting active filter. This phase inversion begins at 180° , suggesting a phase shift of

¶, that is, a phase inversion. This is true at least for low frequencies, below 1 kHz. For frequencies above 1 kHz, there is a phase shift, but it is less than 180° , which is not a strict phase inversion.

Conclusion:

Simulating both circuits yields a gain of 3000, with the difference that in the

The second circuit uses much larger resistors in series to achieve the $30M\Omega$ of R_2 , which is the main resistor of the feedback block; this is not very convenient. On the other hand, the first circuit uses lower value resistors, around 10k to 100k, both in the feedback block and in the resistive divider introduced within the feedback block. This allows for greater stability thanks to the resistive divider and also less variability in resistance values in terms of percentages, since the resistance values are not very high compared to the second circuit where they reach... $M\Omega$.

Exercise #2

Obtain the input impedance using the Laplace Transform for the circuits in Figures 4, 5, and 6.

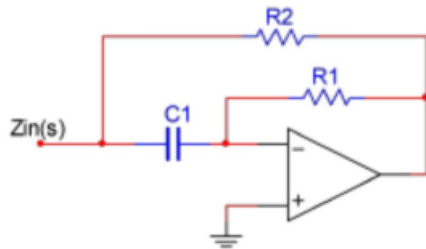


Imagen 4

Circuit of the first dipole to be studied.

Circuit a):

$$Z_{in}(s) = \frac{R_2}{sC_1(R_2+R_1)+1}$$

Ejercicio 2) Se busca obtener la impedancia de entrada usando Laplace

a) $Z_{in}(s)$

- $Z_C(s) = \frac{1}{sC}$
- $Z_R(s) = R$
- $V^- = V^+$ for equilibrium
- Unidad Amplificada

$Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)}$, $I_{in}(s) = I_1(s) + I_2(s)$

con $I_1(s) = I_C(s) = I_{R1}(s)$, $I_2(s) = I_{R2}(s)$

$\Rightarrow I_1(s) = \frac{V_{in}(s) - V^-}{Z_C(s)} = \frac{V_{in}(s) - 0V}{\frac{1}{sC_1}} = sC_1 V_{in}(s)$

$\Rightarrow V_{in}(s) \cdot sC_1 = \frac{V_{in}(s) - V_{out}(s)}{R_1} \Rightarrow V_{in}(s) = \frac{V_{out}(s)}{sC_1 R_1} + V_{in}(s)$

con $A_V(s) = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_2}{R_1} \Rightarrow V_{out}(s) = -V_{in}(s) \cdot \frac{R_2}{R_1}$

$I_2(s) = \frac{V_{in}(s) - V_{out}(s)}{R_2} = \frac{V_{in}(s) - (-V_{in}(s) \cdot \frac{R_2}{R_1})}{R_2} = \frac{V_{in}(s) + V_{in}(s) \cdot \frac{R_2}{R_1}}{R_2}$

$I_{in}(s) = I_1(s) + I_2(s) = \frac{V_{in}(s) - V^-}{Z_C(s)} + \frac{V_{in}(s) - V_{out}(s)}{R_2}$

$I_{in}(s) = \frac{V_{in}(s) - 0V}{\frac{1}{sC_1}} + \frac{V_{in}(s) - (-V_{in}(s) \cdot \frac{R_2}{R_1})}{R_2}$

$I_{in}(s) = V_{in}(s) \cdot sC_1 + \frac{V_{in}(s)}{R_2} + \frac{V_{in}(s) \cdot \frac{R_2}{R_1}}{R_2}$

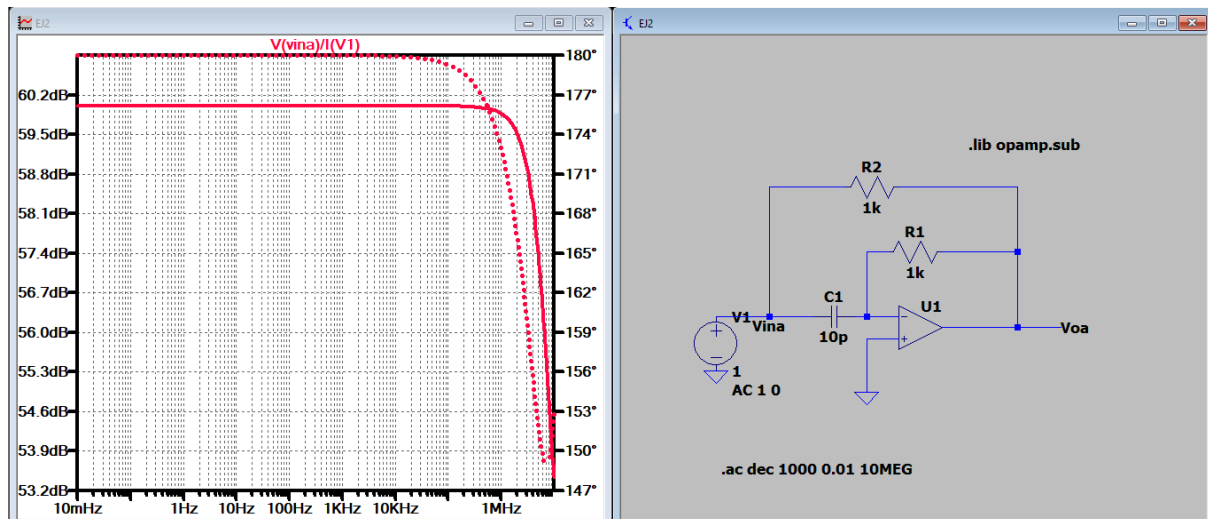
$I_{in}(s) = V_{in}(s) \cdot \left(sC_1 + \frac{1}{R_2} + \frac{sC_1 R_2}{R_1} \right)$

Entonces $\Rightarrow Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)} = \frac{V_{in}(s)}{V_{in}(s) \left(sC_1 + \frac{1}{R_2} + \frac{sC_1 R_2}{R_1} \right)} = \frac{1}{sC_1 + \frac{1}{R_2} + \frac{sC_1 R_2}{R_1}}$

Escaneado con CamScanner

$$Z_{in}(s) = \frac{1}{sC_1 + \frac{1}{\frac{1}{sC_2} + \frac{R_2}{1+sR_2C_2}}} = \frac{R_2}{sC_1(R_2 + 1) + 1} = Z_{in}(s)$$

Simulation:

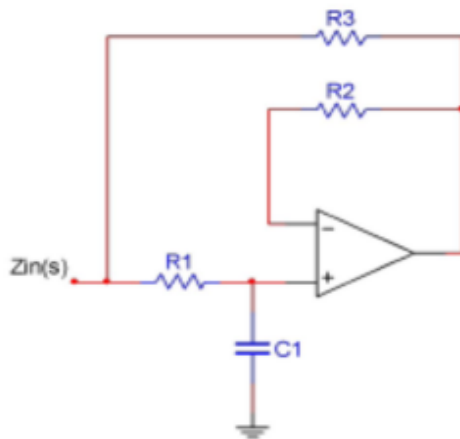


Simulation of the input impedance of the first circuit.

Conclusions The impedance is similar to a lossy capacitor; that is, a capacitor in parallel with an equivalent resistance. $Z_{in}(s)$ depends on both the resistors and the capacitor, components that make up the feedback dipole. The frequency response obtained from the magnitude and phase has this peculiar shape because at low and mid frequencies, since the capacitor's magnitude is so high, the parallel combination of the equivalent resistance and the capacitor ends up giving the equivalent resistance. But as we approach high frequencies, since the capacitor's magnitude decreases considerably, $Z_{in}(s)$ will gradually begin to depend on the capacitor in both magnitude and phase.

It is particularly important to highlight that above a certain frequency, the magnitude and phase change abruptly in a way that does not correspond to the $Z_{in}(s)$ obtained by the nodes. This behavior is due to the fact that the op-amp used for the simulation is not ideal. Therefore, since the Aol and GBW of the op-amp are not infinite, there will be a maximum operating frequency limit at which the op-amp ceases to behave as expected.

Circuit b):



$$Sentence(S) = \frac{R3*(S*C1*R1+1)}{S*C1*(R1+R3)}$$

Imagen 6

Circuit of the second dipole to be studied.

2b)

$$\rightarrow \frac{V_{in} - V_x}{R_1} = \frac{V_x - 0}{\frac{1}{sC_1}}$$

$$\rightarrow \frac{V_o - V_x}{R_2} = 0 \quad \left\{ \begin{array}{l} V_o = V_x \end{array} \right.$$

$$\rightarrow I_{in} = \frac{V_i - V_x}{R_1} + \frac{V_i - V_o}{R_3}$$

$$I_{in} = \frac{V_i - V_o}{R_1} + \frac{V_i - V_o}{R_3}$$

$$I_{in} = V_i \left(\frac{1}{R_1} + \frac{1}{R_3} \right) - V_o \left(\frac{1}{R_1} + \frac{1}{R_3} \right)$$

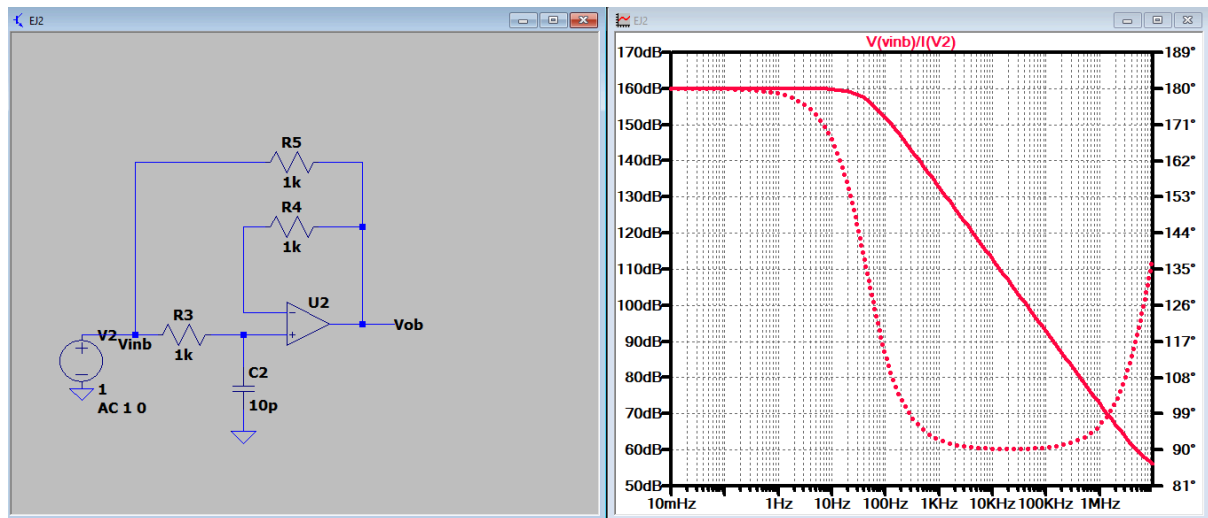
$$I_{in} = V_i \frac{R_3 + R_1}{R_1 R_3} - V_o \frac{R_3 + R_1}{R_1 R_3}$$

$$I_{in} = \frac{V_i (R_3 + R_1)}{R_1 R_3} \left(1 - \frac{1}{sC_1 R_1 + 1} \right) = V_i \frac{R_3 + R_1}{R_1 R_3} \frac{sC_1 R_1}{sC_1 R_1 + 1}$$

$$I_{in} = V_{in} \frac{(R_3 + R_1) \cdot sC_1 R_1}{(sC_1 R_1 + 1) R_1 R_3} = V_{in} \frac{(R_3 + R_1) \cdot sC_1 R_1}{(sC_1 R_1 + 1) R_1 R_3}$$

$$\frac{V_{in}}{I_{in}} = Z_{in} = \frac{(sC_1 R_1 + 1) R_3}{(R_3 + R_1) sC_1} = \frac{sC_1 R_1 R_3 + R_3}{s(C_1 R_1 + C_1 R_3)} = \frac{C_1 R_1 R_3}{C_1 R_1 + C_1 R_3} + \frac{R_3}{s(C_1 R_1 + C_1 R_3)}$$

Simulation:



Simulation of the input impedance of the second circuit.

Conclusions In this case, the impedance obtained at the nodes corresponds to a capacitor in series with a resistor, which depends on capacitor C_2 and resistors R_3 and R_5 (see Figure 7). Looking at the frequency response of $Z_{in}(\omega)$, it can be seen that initially, at low frequencies, the magnitude and phase depend exclusively on the capacitor. Then, starting at a certain frequency, the magnitude stops decreasing, and the phase changes by 90° . This occurs because the magnitude of the resistor in series with the capacitor becomes significant enough that $Z_{in}(\omega)$ depends solely on this resistor, due to the decrease in the capacitor's magnitude as the circuit's operating frequency increases.

It should be clarified that the unusual behavior observed at very low frequencies is due to the limitations of the op-amp used. If the op-amp used for the simulation were more ideal, the module would not be flat at very low frequencies, but would instead follow the characteristic negative slope of the capacitor. And the phase should be at 90° , not 180° .

Circuit c):

$$Sentence(S) = \frac{R1}{S \cdot C1 \cdot (R1 + R2)}$$

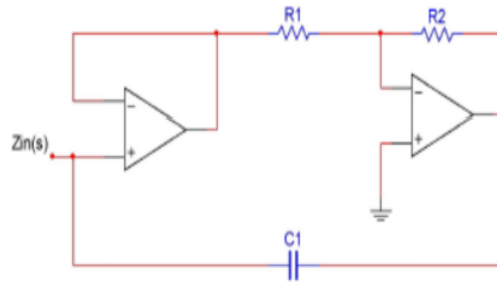


Imagen 8

Circuit of the third

dipole to be studied.

c)

- $V = V^+$ for ogni operazionale en K_1 y K_2 .
- $V_{K1} = V_{K2} = V_{O}$
- $V_{K2} = 0V$
- $Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)}$
- $Z_C(s) = \frac{1}{sC}$
- $Z_R(s) = R$

• $I_{in}(s) = \frac{V_{in}(s) - V_O(s)}{Z_C(s)} = I_1(s)$

con $V^+_{K1} = V_{K1} = V_{in}(s)$, $V^+_{K2} = V_{K2} = 0V$

• $I_2(s) = \frac{V_{K1} - V_{K2}(s)}{Z_{R1}(s)} = \frac{V_{K2} - V_O(s)}{Z_{R2}(s)} \Rightarrow \frac{V_{in}(s) - 0V}{R1} = \frac{0V - V_O(s)}{R2}$

luego $\Rightarrow I_1(s) = \frac{V_{in}(s) - V_O(s)}{1/sC1}$ y reemplazamos $\Rightarrow \frac{V_{in}(s)R2}{R1} = V_O(s)$

• en $\textcircled{2} \Rightarrow I_{in}(s) = I_1(s) = \frac{1}{sC1} [V_{in}(s) - (-V_{in}(s) \frac{R2}{R1})]$

$\Rightarrow I_{in}(s) = V_{in}(s) \cdot \frac{1}{sC1} \left[1 + \frac{R2}{R1} \right]$

$\Rightarrow I_{in}(s) = V_{in}(s) \cdot \frac{1}{sC1} \left(\frac{R1 + R2}{R1} \right)$

Finalmente $\Rightarrow Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)} = \frac{V_{in}(s)}{V_{in}(s) \cdot \frac{1}{sC1} \cdot \frac{(R1 + R2)}{R1}} = \frac{R1}{sC1(R1 + R2)}$

NOTA

Simulation:

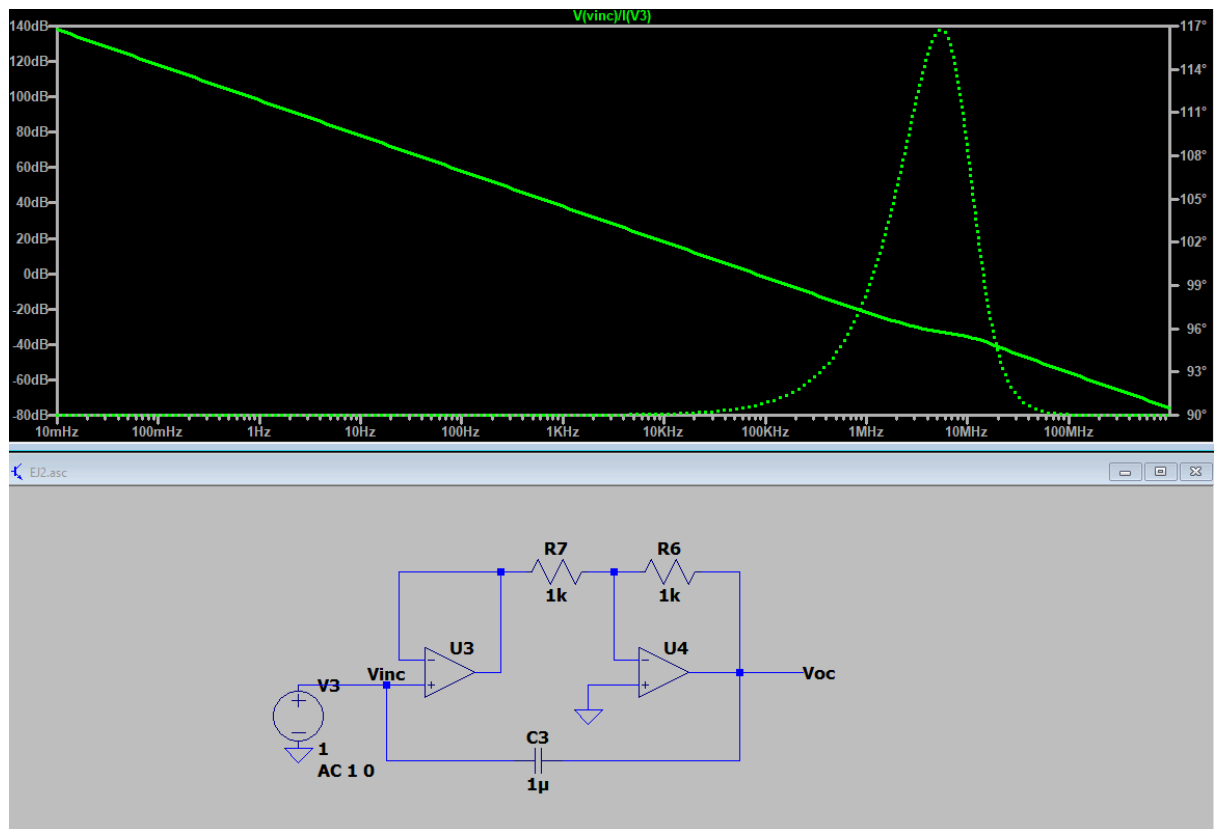


Imagen 9

Simulation of the input impedance of the third circuit.

Conclusions Unlike the previous two, its impedance is similar to that of a lossless capacitor.

Exercise #3

We have an amplifier stage with differential output like the one in image 10, we want $V_o(s) = V_A(s) - V_B(s)$

with $Day = \frac{R^*(k-1)}{2}$ given by statement.

$$V_A - V_B = V_1 * 2k$$

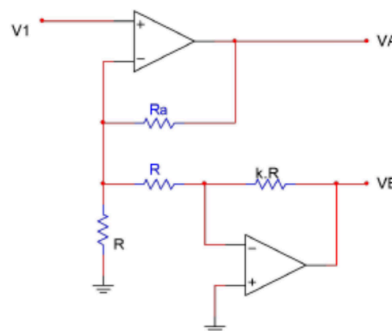


Imagen 10

Circuit of the amplifier with differential output to be studied.

Handwritten solution for Exercise #3:

$V_{in}(s) = V_1(s)$

Assume amplification ideal $\Rightarrow V_{K1} = V_{K1}^+$ y $V_{K2} = V_{K2}^+$ por equipotencialidad.

De amplificador $K_1 \Rightarrow I_1(s) = \frac{V_{in}(s) - V_A(s)}{Z_{K1}(s)} = -\frac{V_{in}(s) - V_A(s)}{R_2} = I_2(s)$

donde $I_1(s) = I_2(s) + I_3(s)$, $\frac{V_{in}(s) - V_{K2}}{Z_{K1}(s)} = \frac{V_{K2} - 0V}{Z_{K2}(s)} = I_2(s)$

con $I_3(s) = \frac{V_{in}(s) - 0V}{Z_{K3}(s)}$

luego simplifiquemos $I_2(s)$ y $I_3(s)$ en $I_1(s)$, así queda como:

$$I_1(s) = I_2(s) + I_3(s) \Rightarrow -\frac{V_{in}(s) - V_A(s)}{R_2} = \frac{V_{in}(s) - 0V}{R_2} + \frac{V_{in}(s) - 0V}{R}$$

$$\Rightarrow -\frac{V_{in}(s) - V_A(s)}{R_2} = +\frac{V_{in}(s)}{R_2} + \frac{V_{in}(s)}{R}$$

$$\Rightarrow -\frac{V_{in}(s)}{R_2} + \frac{V_A(s)}{R_2} = +\frac{V_{in}(s)}{R_2} + \frac{V_{in}(s)}{R}$$

$$V_A(s) = \left(2 \cdot \frac{R_2}{R} + 1\right) V_{in}(s)$$

De I_1 y $I_2 \Rightarrow \frac{V_{in}(s) - 0V}{R} = \frac{0V - V_B(s)}{K \cdot R} \Rightarrow -V_{in}(s) \cdot K = V_B(s)$

Entonces:

$$V_A(s) - V_B(s) = V_{in}(s) \left(2 \cdot \frac{R_2}{R} + 1\right) - (-V_{in}(s) \cdot K)$$

$$V_A(s) - V_B(s) = V_{in}(s) \left[2 \cdot \frac{R_2}{R} + 1 + K\right]$$

$$V_A(s) - V_B(s) = V_{in}(s) \left[\frac{2 \cdot R_2(K-1)}{R} + 1 + K\right]$$

$$V_A(s) - V_B(s) = V_{in}(s) [K-1+1+K]$$

$$V_A(s) - V_B(s) = 2K \cdot V_{in}(s)$$

Simulation:

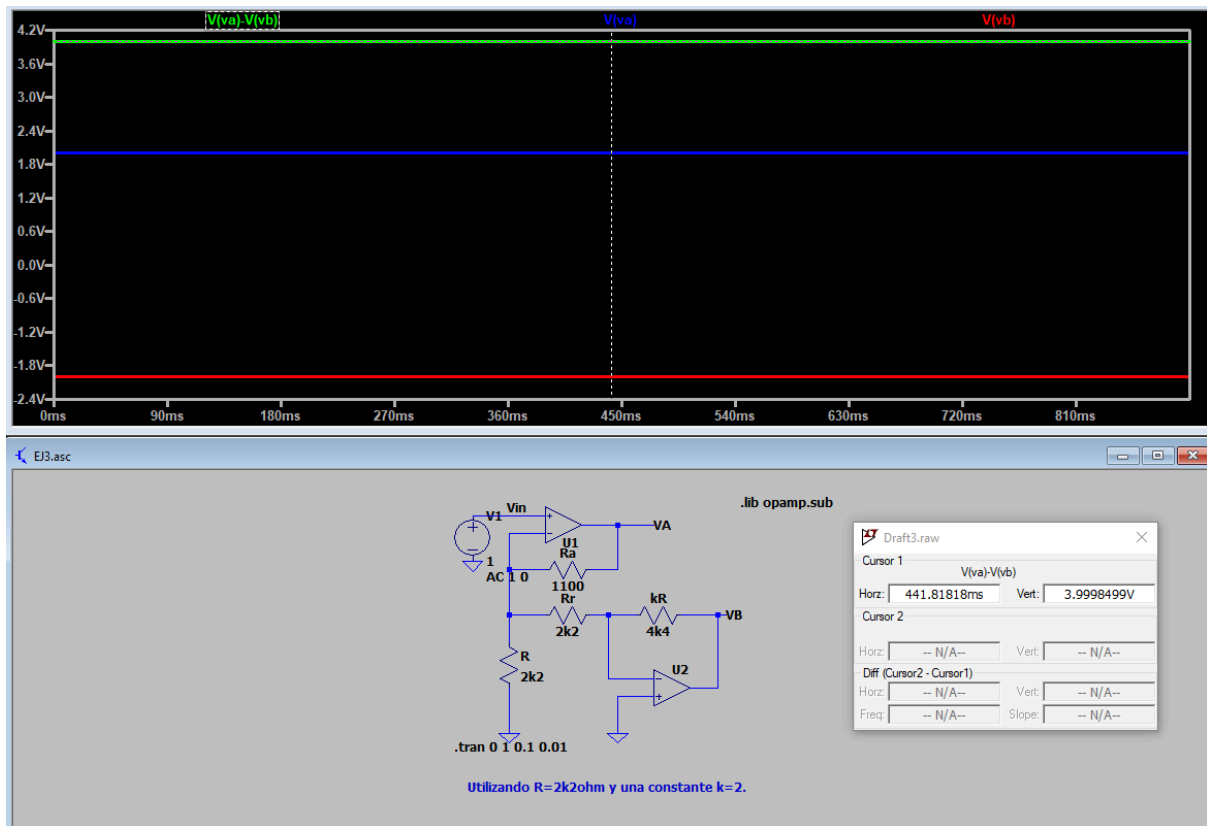


Imagen 11

Simulation of the difference between potentials VA and VB.

Exercise #4

Obtain the transfer function V_2/V_1 (magnitude, phase, and pole-zero diagram). Given $R_2/R_1=1$ and $R_A/R_B=5$. Simulate with $R=R_3=1k\Omega$ and $C=1\mu F$.

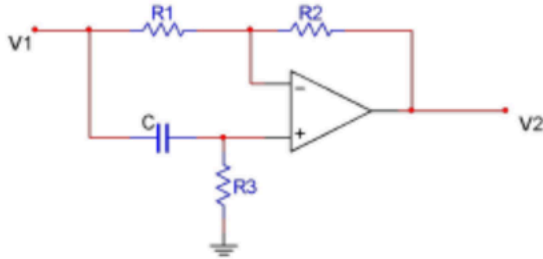


Imagen 12

First phase-rotating circuit to be studied.

Circuit a) We have an All-Pass Filter or a Phase Rotator, we want $H(S) = \frac{V_2(S)}{V_1(S)}$. We took the module, the phase, and the pole-zero diagram. $\frac{R_2}{R_1} = 1$.

$$H(S) = \frac{S * C * R_3 - 1}{S * C * R_3 + 1}$$

$$H(S) = \frac{S - 1000}{S + 1000} \text{ where } R_3 = 1k\Omega, C = 1\mu F$$

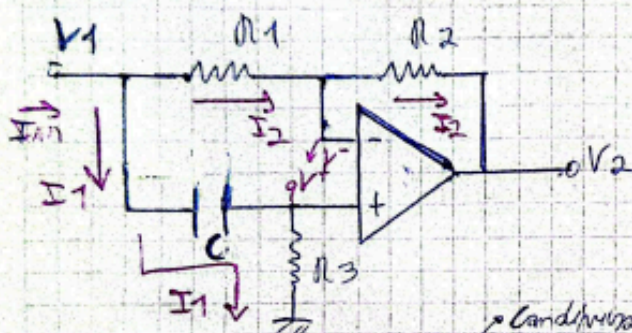
$$|H(j\omega)| = 1$$

$$\phi(\omega) = -2 \arctan(\omega * C * R_3)$$

Ejercicio 4)

a) Filtro Paso Bajas o Integrador de Fase, queremos $H(s) = \frac{V_2(s)}{V_1(s)}$ tanto el módulo, la fase y el diagrama de Bode y los ceros.

• Queremos $\frac{R_2}{R_1} = 1$ y $\frac{R_1}{R_3} = 5$



- Asumo filtros ideales
- $V^+ = V^-$ por equipotencialidad
- $Y_C(s) = sC$, $Y_{R1}(s) = \frac{1}{R_1}$

• $V^+ = V^- = V_1 \cdot \frac{Y_C(s)}{Y_C(s) + Y_{R3}(s)}$

• Condición de tensión
 $I_{in}(s) = I_1(s) + I_2(s)$

• $I_1(s) = (V_1(s) - V^+) \cdot Y_C(s) = (V^+ - 0V) \cdot Y_{R3}(s) \Rightarrow V_1(s) \cdot Y_C(s) - V^+ \cdot Y_C(s) = V^+ \cdot Y_{R3}(s)$

$\Rightarrow V_1(s) \cdot Y_C(s) = V^+ \cdot Y_C(s) + V^+ \cdot Y_{R3}(s) \Rightarrow V_1(s) \cdot Y_C(s) = V^+ (Y_C(s) + Y_{R3}(s))$

$\Rightarrow \frac{V_1(s) \cdot Y_C(s)}{Y_C(s) + Y_{R3}(s)} = V^+(s) \rightarrow$ por igualdad de conductancias.

• $I_2(s) = (V_1(s) - V^-) \cdot Y_{R1}(s) = (V^- - V_2(s)) \cdot Y_{R2}(s)$

$\Rightarrow I_2(s) = V_1(s) \cdot Y_{R1}(s) - V^- \cdot Y_{R1}(s) = V^- \cdot Y_{R2}(s) - V_2(s) \cdot Y_{R2}(s)$

$\Rightarrow V_1(s) \cdot Y_{R1}(s) + V_2(s) \cdot Y_{R2}(s) = V^- \cdot Y_{R2}(s) + V^- \cdot Y_{R1}(s)$

$V_1(s) \cdot Y_{R1}(s) + V_2(s) \cdot Y_{R2}(s) = V^- (Y_{R2}(s) + Y_{R1}(s))$, $V^- = V^+$

$V_1(s) \cdot Y_{R1}(s) + V_2(s) \cdot Y_{R2}(s) = V_1(s) \cdot Y_C(s) \cdot \left(\frac{Y_{R2}(s) + Y_{R1}(s)}{Y_C(s) + Y_{R3}(s)} \right)$

$V_2(s) \cdot Y_{R2}(s) = V_1(s) \cdot \left(\frac{Y_C(s) \cdot Y_{R2}(s)}{Y_C(s) + Y_{R3}(s)} + \frac{Y_{R1}(s) \cdot Y_{R2}(s)}{Y_C(s) + Y_{R3}(s)} \right) \cdot \frac{1}{Y_{R2}(s)}$

$\Rightarrow \frac{V_2(s)}{V_1(s)} = \frac{Y_C(s) \cdot Y_{R2}(s)}{Y_C(s) + Y_{R3}(s)} \cdot \frac{1}{Y_{R2}(s)} + \frac{Y_{R1}(s) \cdot Y_{R2}(s)}{Y_C(s) + Y_{R3}(s)} \cdot \frac{1}{Y_{R2}(s)} = \frac{Y_{R1}(s)}{Y_{R2}(s)}$

$$\frac{V_2(s)}{V_1(s)} = \frac{V_c(s)}{V_c(s) + V_{n2}(s)} \cdot \left(1 + \frac{Y_{n1}(s)}{Y_{n2}(s)} \right) - \frac{V_{n1}(s)}{V_{n2}(s)}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{g_c}{g_c + \frac{1}{n_3}} \cdot \left(1 + \frac{1/n_1}{1/n_2} \right) - \frac{1/n_1}{1/n_2}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{g_c n_3}{(g_c + \frac{1}{n_3}) n_3} \left(1 + \frac{n_2}{n_1} \right) - \frac{n_2}{n_1}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{g_c n_3}{g_c n_3 + 1} \left(1 + \frac{n_2}{n_1} \right) - \frac{n_2}{n_1} \quad (A)$$

$$H(s) = \frac{g_c n_3 n_1 \left(1 + \frac{n_2}{n_1} \right) - (g_c n_3 + 1) n_2}{(g_c n_3 + 1) \cdot n_1}$$

$$H(s) = \frac{g_c n_3 n_1 + g_c n_3 n_2 - g_c n_3 n_2 - n_2}{g_c n_3 n_1 + 1}$$

$$H(s) = \frac{g_c n_3 n_1 - n_2}{g_c n_3 n_1 + 1} \rightarrow \text{Transferencia, expresión general} \quad (B)$$

Partiendo de (B) $\Rightarrow H(s) = \frac{g_c n_3 (1+1) - 1}{g_c n_3 + 1}$

$$H(s) = \frac{2 \cdot g_c n_3}{g_c n_3 + 1} - 1$$

$$H(s) = \frac{2 g_c n_3 - (g_c n_3 + 1)}{g_c n_3 + 1}$$

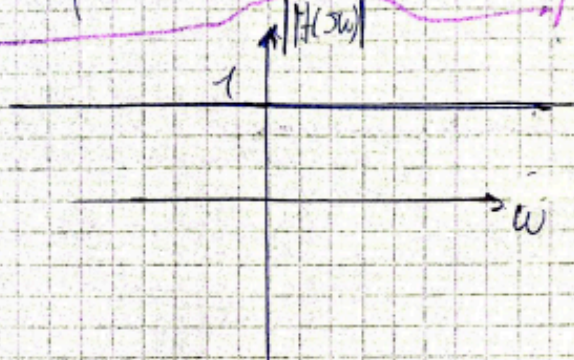
$$H(s) = \frac{2 \cdot g_c n_3 - g_c n_3 - 1}{g_c n_3 + 1}$$

$$H(s) = \frac{g_c n_3 - 1}{g_c n_3 + 1} \rightarrow \text{con } \frac{n_2}{n_1} = 1$$

Buscando el módulo

$$\Phi = 5\omega \Rightarrow H(5\omega) = \frac{5\omega \cdot c \cdot \omega^3 - 1}{5\omega \cdot (c \cdot \omega^3 + 1)} \Rightarrow |H(5\omega)| = \frac{\sqrt{(-1)^2 + (\omega \cdot c \cdot \omega^3)^2}}{\sqrt{1^2 + (\omega \cdot c \cdot \omega^3)^2}}$$

$$\Rightarrow |H(5\omega)| = 1 \rightarrow \text{Es un paso todo} \rightarrow \text{módulo}$$



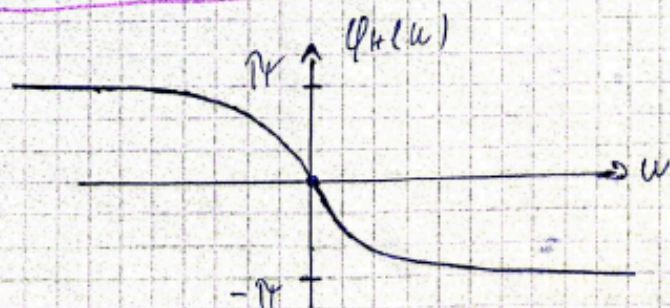
Buscando la fase

$$\varphi_H(\omega) = \arctan\left(\frac{\text{Im}\{N_{\text{num}} H(5\omega)\}}{\text{Re}\{N_{\text{num}} H(5\omega)\}}\right) - \arctan\left(\frac{\text{Im}\{D_{\text{den}} H(5\omega)\}}{\text{Re}\{D_{\text{den}} H(5\omega)\}}\right)$$

$$\varphi_H(\omega) = \arctan\left(\frac{\omega \cdot c \cdot \omega^3}{-1}\right) - \arctan\left(\frac{\omega \cdot c \cdot \omega^3}{1}\right)$$

$$\varphi_H(\omega) = -\arctan(\omega \cdot c \cdot \omega^3) - \arctan(\omega \cdot c \cdot \omega^3)$$

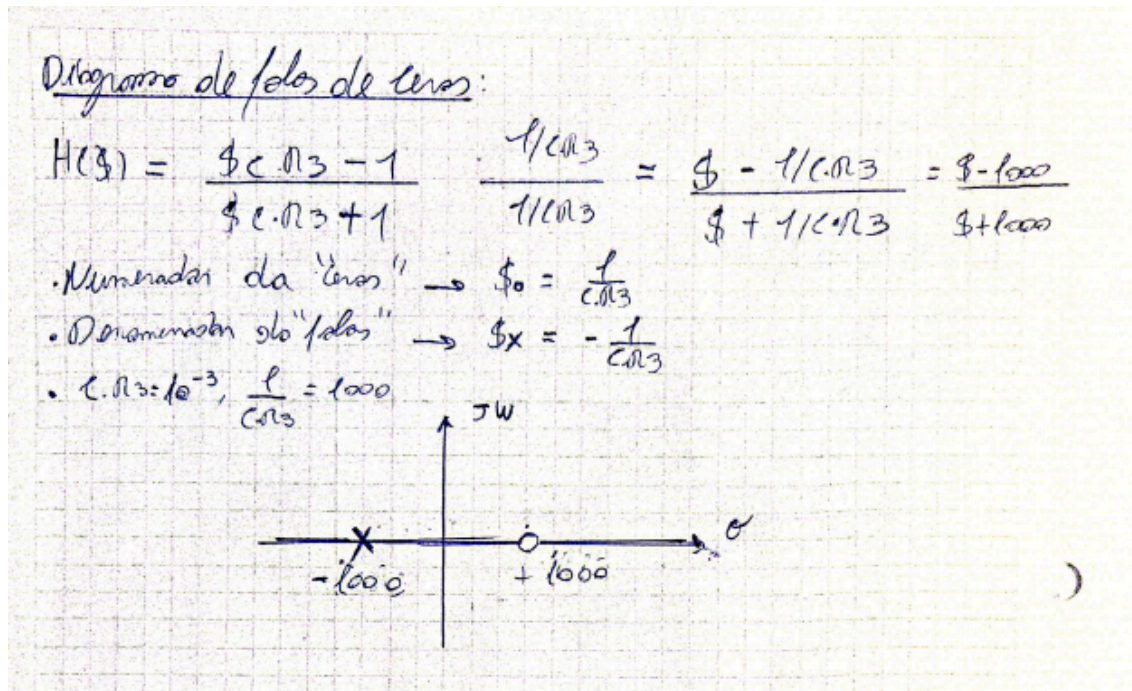
$$\varphi_H(\omega) = -2 \cdot \arctan(\omega \cdot c \cdot \omega^3) \rightarrow \text{fase}$$



$$\bullet \omega = 0 \Rightarrow \varphi_H(0) = 0$$

$$\bullet \omega \rightarrow +\infty \Rightarrow \varphi_H(\omega) \rightarrow -\pi$$

$$\bullet \omega \rightarrow -\infty \Rightarrow \varphi_H(\omega) \rightarrow \pi$$



Simulation:

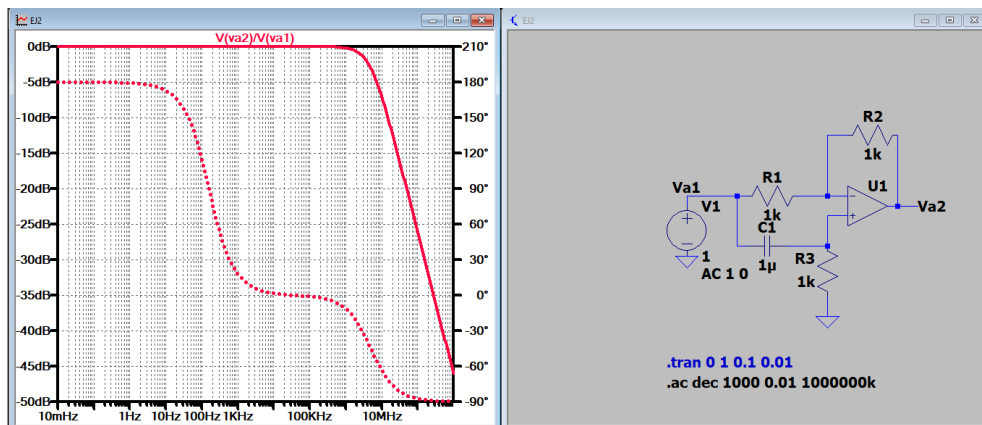


Imagen 13

Transfer function simulation of the first circuit.

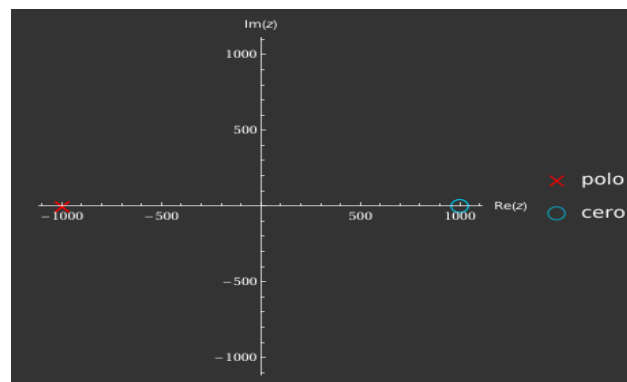


Imagen 14

Simulation of the poles and zeros of the first circuit.

Dotted Line → Phase

Full Line → Module

Conclusion In this first circuit, it is mentioned that it is an All-Pass Filter or Phase Rotator, meaning that the transfer factor is 1. This results in any input signal not being attenuated at the output in terms of amplitude over a large frequency range. That is, the product of the signal's amplitude and the transfer factor equals the signal's amplitude.

However, a 180° phase rotation occurs. This is observed in the expression for the signal phase, which is: $-2 \cdot \arctan(\omega R_3 C)$ While graphically in LTSpice, we observe that at low frequencies the phase is 180° , but at high frequencies (1 MHz) the phase decreases to 0° . Then, it rotates back, beginning to increase negatively for frequencies above 1 MHz, reaching approximately -90° . This is because, since the op-amp has an internal pole, from this frequency onward it begins to attenuate the transfer modulus, introducing a -90° phase shift.

On the other hand, if we look at the graph of the phase performed without a simulator, we see that:

- For low frequencies (ω close to 0), the phase angle is almost 0° .
- When $\omega R C = 1$, the phase angle is 90° .
- For high positive frequencies ($\omega \rightarrow \infty$), the phase angle approaches -180° .
- For high negative frequencies ($\omega \rightarrow -\infty$), the phase angle approaches $+180^\circ$.

The negative sign indicates that the output signal is out of phase with the input signal. In other words, the circuit introduces a time delay as the frequency increases.

"Rotation" refers to the advance or delay of the signal in time, which in the frequency domain translates to a change in phase angle. This phase shift is useful in applications such as power factor correction, oscillation generation, or delay compensation in control systems.

In summary, an op-amp-based all-pass filter or phase rotator works by modifying the phase of the input signal using an RC network, leaving the gain virtually unchanged. Phase control is achieved through the RC components connected to the op-amp's positive terminal.

Circuit b) We have an All-Pass Filter or a Phase Rotator, we want $H(s) = \frac{V_2(s)}{V_1(s)}$ We took the module, the phase, and the pole-zero diagram. $\frac{SUN}{RB} = 5$.

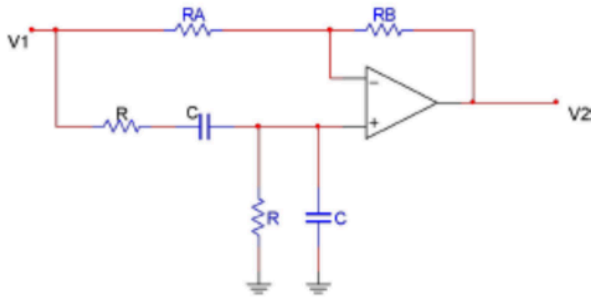


Imagen 15

Second phased rotary circuit to be studied.

$$H(S) = (1/5) * \frac{-S^2 + S * \frac{3}{C*R} - \frac{1}{(C*R)^2}}{S^2 + S * \frac{3}{C*R} + \frac{1}{(C*R)^2}}$$

$$H(S) = (1/5) * \frac{(S-2618) * (S-381,96)}{(S+2618) * (S+381,96)} \text{ where } R = 1\text{kohm}, C = 1\mu\text{F}$$

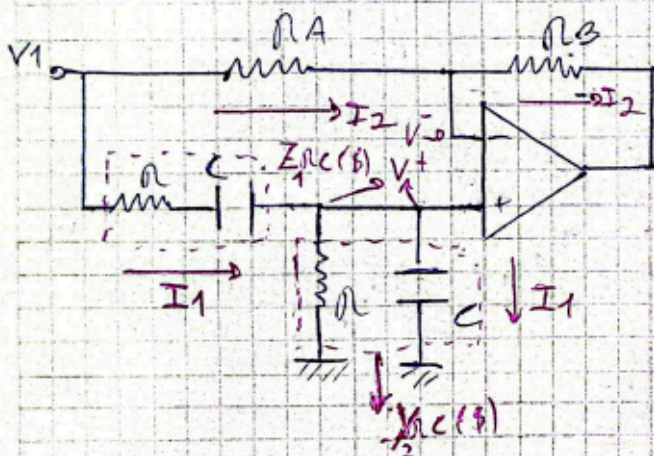
$$|H(j\omega)| = 1/5$$

$$\phi_H(\omega) = -2 \arctg\left(\frac{3 \text{ hours}}{-\omega^2 * C * R + \frac{1}{C * R}}\right)$$

Ejercicio 4)

b)

- Filtro pasa todo o integrador de Foxe, queremos $H(s) = \frac{V_2(s)}{V_1(s)}$ tanto el módulo, la fase, y el diagrama de polos y ceros
- Tomo $\frac{R_2}{R_1} = 1$ y $\frac{R_A}{R_B} = 5$



• Asumo filtro ideal

• $Z_R(s) = R$, $Z_C(s) = \frac{1}{sC}$

• $Y_R(s) = \frac{1}{R}$, $Y_C(s) = sC$

• $V^+ = V^-$ por equipotencialidad

$$V^+ = V^- = V_1(s) \frac{Y_{RC}(s)}{Y_{1RC}(s) + Y_{2RC}(s)}, \quad I_{in}(s) = I_1(s) + I_2(s)$$

• $I_1(s) = (V_1(s) - V^+) \cdot Y_{1RC}(s) = (V^+ - 0V) \cdot Y_{2RC}(s)$ → Por equipotencialidad de bornes y análogamente al ejercicio 4) $\Rightarrow$ se obtiene

$$V^+ = V_1(s) \frac{Y_{1RC}(s)}{Y_{1RC}(s) + Y_{2RC}(s)} \quad \text{con } V^+ = V^-$$

$$I_2(s) = (V_1(s) - V^-) \cdot Y_{RA}(s) = (V^- - V_2(s)) \cdot Y_{RB}(s)$$

$$\Rightarrow V_1(s) \cdot Y_{RA}(s) - V^- \cdot Y_{RA}(s) = V^- \cdot Y_{RB}(s) - V_2(s) \cdot Y_{RB}(s)$$

$$\Rightarrow V_1(s) \cdot Y_{RA}(s) + V_2(s) \cdot Y_{RB}(s) = V^- \cdot Y_{RB}(s) + V^- \cdot Y_{RA}(s)$$

$$V_1(s) \cdot Y_{RA}(s) + V_2(s) \cdot Y_{RB}(s) = V^- \cdot (Y_{RB}(s) + Y_{RA}(s))$$

$$V_1(s) \cdot Y_{NA}(s) + V_2(s) \cdot X_{AB}(s) = V_1(s) \cdot \frac{Y_{AC}(s)}{Y_{AC}(s) + Y_{BC}(s)} \cdot (X_{AB}(s) + Y_{AC}(s))$$

$$V_2(s) \cdot X_{AB}(s) = V_1(s) \left(\frac{Y_{BC}(s) \cdot X_{AB}(s)}{Y_{AC}(s) + Y_{BC}(s)} + \frac{Y_{AC}(s) \cdot Y_{NA}(s)}{Y_{AC}(s) + Y_{BC}(s)} \right) - V_1(s) \cdot Y_{AC}(s)$$

$$\frac{V_2(s)}{V_1(s)} = \frac{Y_{BC}(s) \cdot \cancel{X_{AB}(s)} \cdot 1}{Y_{AC}(s) + Y_{BC}(s) \cdot \cancel{X_{AB}(s)}} + \frac{Y_{AC}(s) \cdot Y_{NA}(s) \cdot 1}{Y_{AC}(s) + Y_{BC}(s) \cdot X_{AB}(s)} - \frac{Y_{AC}(s)}{Y_{AB}(s)}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{Y_{BC}(s)}{Y_{AC}(s) + Y_{BC}(s)} \left(1 + \frac{Y_{NA}(s)}{Y_{AB}(s)} \right) - \frac{Y_{AC}(s)}{Y_{AB}(s)}$$

$$Y_{AC}(s) = \frac{1}{Z_{AC}(s)} = \frac{1}{R + \frac{1}{sC}} = \frac{sC}{sCR + 1}$$

$$Y_{BC}(s) = \frac{1}{Z_{BC}(s)} = Y_{AB}(s) + Y_{CC}(s) = \frac{1}{R} + sC = \frac{1 + sCR}{R} = \frac{sCR + 1}{R}$$

$$Y_{NA}(s) = \frac{1}{R_A}, \quad Y_{AB}(s) = \frac{1}{R_B}$$

$$\frac{Y_{AC}(s)}{Y_{AC}(s) + Y_{BC}(s)} = \frac{\frac{sC}{sCR + 1}}{\frac{sC}{sCR + 1} + \frac{sCR + 1}{R}} = \frac{sC}{(sCR + 1) \left[\frac{sC}{sCR + 1} + \frac{sCR + 1}{R} \right]}$$

$$= \frac{sC \cdot R}{sC + (sCR + 1)^2} \cdot \frac{R}{R} = \frac{sCR}{sCR + (sCR + 1)^2} = \frac{sCR}{s^2 CR^2 + 3sCR + 1}$$

$$= \frac{sCR}{s^2 CR^2 + 3sCR + 1} \cdot \frac{1/(CR)^2}{1/(CR)^2} = \boxed{\frac{s/CR}{s^2 + 3s \frac{1}{CR} + \frac{1}{(CR)^2}} = \frac{Y_{AC}(s)}{Y_{BC}(s) + Y_{AC}(s)}}$$

$$\frac{Y_{NA}(s)}{Y_{AB}(s)} = \frac{1/R_A}{1/R_B} = \frac{R_B}{R_A}$$

$$H(\xi) = \frac{\xi/cn}{\xi^2 + 3\xi \frac{1}{cn} + \frac{1}{(cn)^2}} \left(1 + \frac{nb}{nA} \right) - \frac{nb}{nA} \quad \textcircled{4}$$

$$H(\xi) = \frac{\frac{\xi \cdot nA}{cn} \left(1 + \frac{nb}{nA} \right) - nb \cdot \left(\xi^2 + \xi \cdot \frac{3}{cn} + \frac{1}{(cn)^2} \right)}{\left(\xi^2 + \xi \cdot \frac{3}{cn} + \frac{1}{(cn)^2} \right) \cdot nA}$$

$$H(\xi) = \frac{\xi \cdot \frac{nA}{cn} + \frac{\xi nb}{cn} - \xi^2 nb - \xi \cdot \frac{3nb}{cn} - \frac{nb}{(cn)^2}}{\xi^2 \cdot nA + \xi \cdot \frac{3 \cdot nA}{cn} + \frac{nA}{(cn)^2}}$$

$$H(\xi) = \frac{-\xi^2 nb - \xi \cdot \frac{2nb}{cn} + \xi \cdot \frac{nA}{cn} - \frac{nb}{(cn)^2}}{\xi^2 \cdot nA + \xi \cdot \frac{3 \cdot nA}{cn} + \frac{nA}{(cn)^2}}$$

Transfórmica, expres. general

• Ahora aplica la relación $\frac{nA}{nb} = 5 \Leftrightarrow \frac{nb}{nA} = \frac{1}{5}$

$$\Rightarrow \text{Retomando } \textcircled{4} \Rightarrow H(\xi) = \frac{\frac{\xi}{cn}}{\xi^2 + 3\xi \frac{1}{cn} + \frac{1}{(cn)^2}} \left(1 + \frac{1}{5} \right) - \frac{1}{5}$$

$$H(\xi) = \frac{\frac{\xi}{cn} \cdot \frac{6}{5} \cdot 5 - \left(\xi^2 + 3\xi \frac{1}{cn} + \frac{1}{(cn)^2} \right)}{\left(\xi^2 + 3\xi \frac{1}{cn} + \frac{1}{(cn)^2} \right) \cdot 5}$$

$$H(\xi) = \frac{-\xi^2 - 3\xi \frac{1}{cn} + 6\xi \frac{1}{cn} - \frac{1}{(cn)^2}}{\xi^2 + 3 \cdot \frac{1}{cn} + \frac{1}{(cn)^2}} \cdot \frac{1}{5} \quad K = \frac{1}{5}$$

$$H(\xi) = \frac{-\xi^2 + 3 \frac{1}{cn} - \frac{1}{(cn)^2}}{\xi^2 + 3 \frac{1}{cn} + \frac{1}{(cn)^2}} \cdot \frac{1}{5} \quad \rightarrow \lim_{\xi \rightarrow 0} \frac{nA}{nb} = 5$$

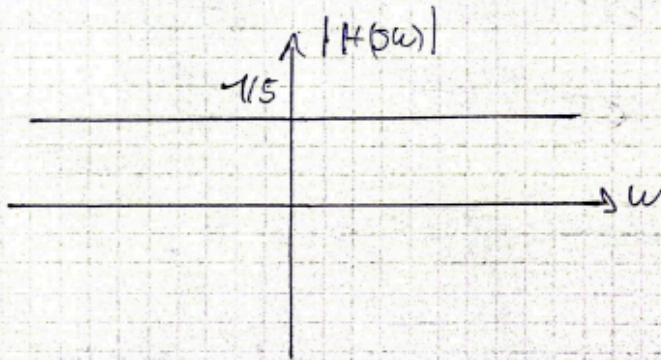
Busca el módulo

$$s = j\omega \Rightarrow H(s\omega) = \frac{+ (j\omega)^2 - 3 \cdot \frac{j\omega}{c.n} + \frac{1}{(c.n)^2}}{(j\omega)^2 + 3 \cdot \frac{j\omega}{c.n} + \frac{1}{(c.n)^2}} \cdot \frac{1 \cdot (-1)}{5}$$

$$H(j\omega) = \frac{-\omega^2 + j \cdot \frac{3\omega}{c.n} + \frac{1}{(c.n)^2}}{-\omega^2 + j \frac{3\omega}{c.n} + \frac{1}{(c.n)^2}} \cdot \frac{1}{5}$$

$$|H(j\omega)| = \frac{1}{5} \cdot \frac{\sqrt{(\frac{1}{(c.n)^2} - \omega^2)^2 + (\frac{3\omega}{c.n})^2}}{\sqrt{(\frac{1}{(c.n)^2} - \omega^2)^2 + (\frac{3\omega}{c.n})^2}} = \frac{1}{5} = |H(j\omega)|$$

• Por lo tanto el módulo es $\frac{1}{5}$



Busca la fase:

$$\phi_H(\omega) = \arctg\left(\frac{\text{Im}\{ \text{Num } H(s\omega) \}}{\text{Re}\{ \text{Num } H(s\omega) \}}\right) - \arctg\left(\frac{\text{Im}\{ \text{Den } H(s\omega) \}}{\text{Re}\{ \text{Den } H(s\omega) \}}\right)$$

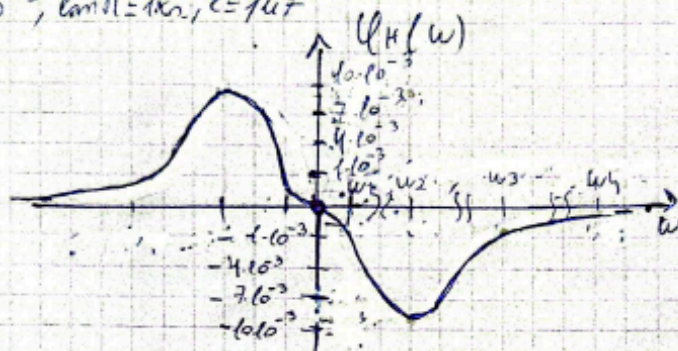
$$\phi_H(\omega) = \arctg\left(\frac{-\frac{3\omega}{c.n}}{\frac{1}{(c.n)^2} - \omega^2}\right) - \arctg\left(\frac{\frac{3\omega}{c.n}}{\frac{1}{(c.n)^2} - \omega^2}\right)$$

$$\phi_H(\omega) = -\arctg\left(\frac{\frac{3\omega}{c.n}}{\frac{1}{(c.n)^2} - \omega^2}\right) - \arctg\left(\frac{-\frac{3\omega}{c.n}}{\frac{1}{(c.n)^2} - \omega^2}\right)$$

$$\phi_H(\omega) = -2 \arctg\left(\frac{\frac{3\omega}{c.n}}{\frac{1}{(c.n)^2} - \omega^2}\right)$$

$$\varphi_H(\omega) = -2 \arctan \left(\frac{\frac{3\omega}{c \cdot R}}{\frac{1}{(c \cdot R)^2} - \omega^2} \right) = -2 \arctan \left(\frac{3\omega}{-\omega^2 c R + \frac{1}{c \cdot R}} \right)$$

$$c \cdot R = 10^{-3}, \text{ em } R = 1 \text{ k}\Omega, c = 1 \mu\text{F}$$



$$\omega = 0 \Rightarrow \varphi_H(0) = 0$$

$$\omega \rightarrow +\infty \Rightarrow \varphi_H(\omega) \rightarrow 0$$

$$\omega \rightarrow -\infty \Rightarrow \varphi_H(\omega) \rightarrow 0$$

- $f_1 = 10 \text{ kHz} \Rightarrow \omega_1 = 2\pi \cdot 10^4 \text{ rad/s}$
- $f_2 = 100 \text{ kHz} \Rightarrow \omega_2 = 2\pi \cdot 10^5 \text{ rad/s}$
- $f_3 = 1 \text{ MHz} \Rightarrow \omega_3 = 2\pi \cdot 10^6 \text{ rad/s}$
- $f_4 = 10 \text{ MHz} \Rightarrow \omega_4 = 2\pi \cdot 10^7 \text{ rad/s}$

$\varphi_H(\omega)$ rad	f
0	0
$\approx \pm 0.37 \cdot 10^{-3}$	$\pm 10 \text{ kHz}$
$\approx \pm 0.6 \cdot 10^{-3}$	$\pm 100 \text{ kHz}$
$\approx \pm 0.95 \cdot 10^{-3}$	$\pm 1 \text{ MHz}$
$\approx \pm 0.995 \cdot 10^{-3}$	$\pm 10 \text{ MHz}$

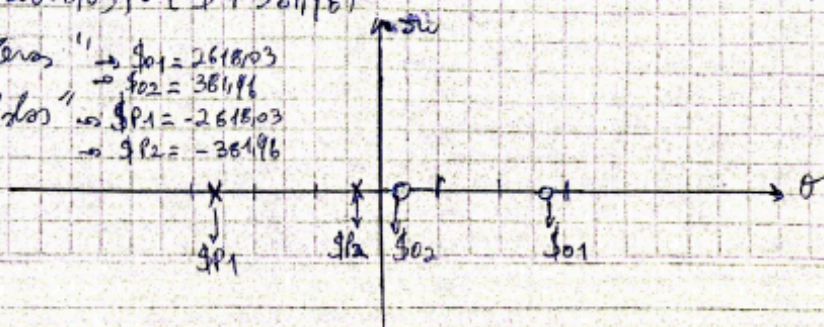
Diagrama de polos e zeros

$$H(s) = \frac{-s^2 + 3 \frac{s}{c \cdot R} - \frac{1}{(c \cdot R)^2}}{s^2 + 3 \frac{s}{c \cdot R} + \frac{1}{(c \cdot R)^2}} \cdot \frac{1}{5}$$

$$\text{Se } c \cdot R = 10^{-3} \quad \text{em } R = 1 \text{ k}\Omega, c = 1 \mu\text{F}$$

$$H(s) = \frac{(s - 2618.03) \cdot (s - 381.966)}{(s + 2618.03) \cdot (s + 381.966)} \cdot \frac{1}{5}$$

- Numerador do "Zero" $\rightarrow s_{01} = 2618.03$
 $\rightarrow s_{02} = 381.96$
- Denominador do "Polo" $\rightarrow s_{p1} = -2618.03$
 $\rightarrow s_{p2} = -381.96$



NOTA

Simulation:

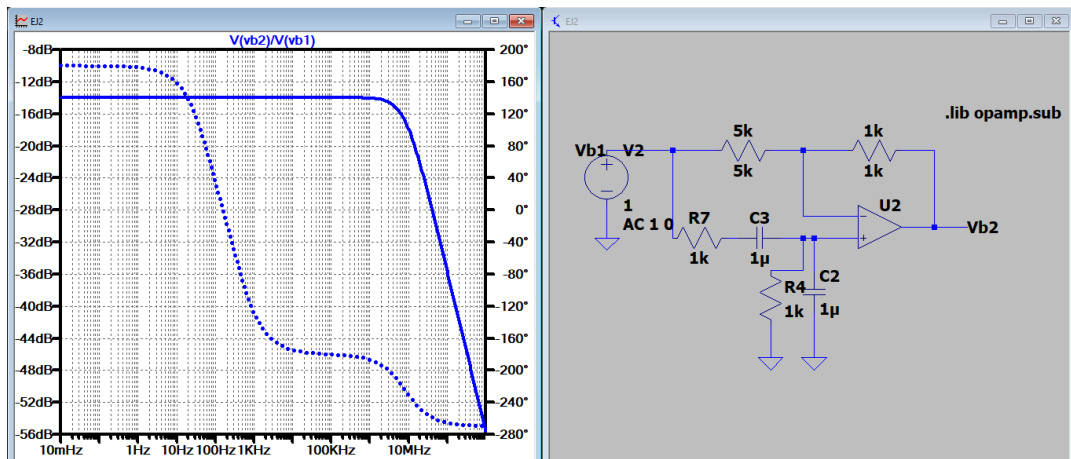


Imagen 16

Simulation of the transfer function of the second circuit.

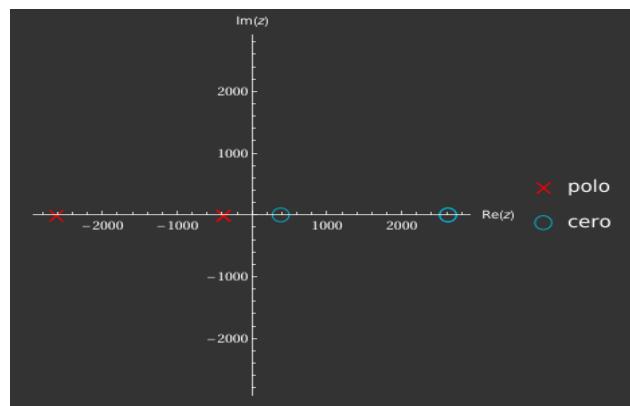


Imagen 17

Simulation of the poles and zeros of the second circuit.

Dotted Line → Phase

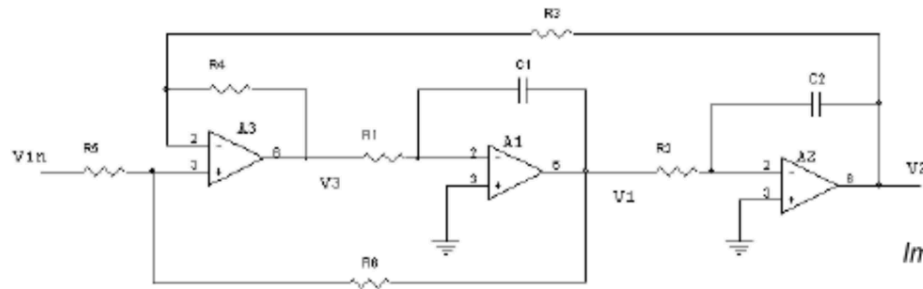
Full Line → Module

Conclusion In this second circuit, we still have an all-pass filter or phase rotator, but with the peculiarity that there is an attenuation of approximately a factor of $1/5$, meaning about -14 dB across the entire module. This is because the ratio between resistors R_A and R_B is not 1. Therefore, since R_A is five times greater than R_B , the output signal will be five times smaller than the input signal.

Furthermore, due to the presence of two reactive elements in the circuit, the transfer function consists of two zeros and two poles. This results in a phase shift of $\pm 180^\circ$ depending on the signal frequency. Specifically, at low frequencies, the output signal will lead the input signal by 180° , and at high frequencies, the output signal will lag the input signal by 180° .

Exercise #5

Obtain the transfers and compare them for the next state variable filter.



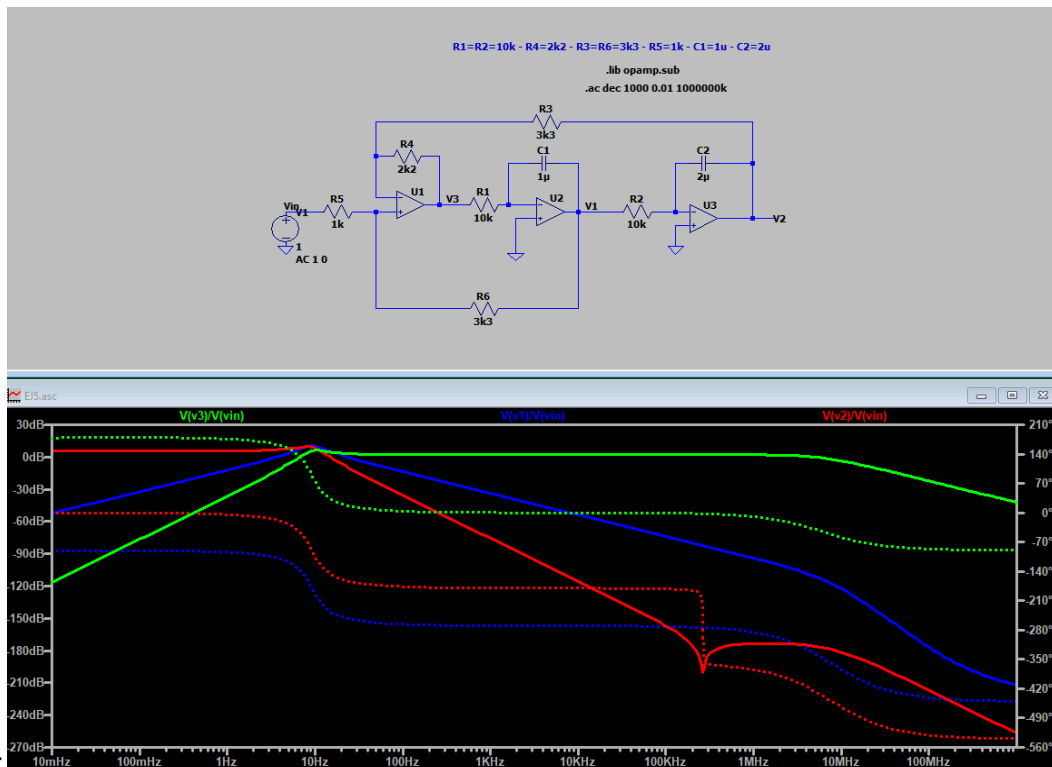
Filter of state variables to be analyzed.

$$\frac{V3(S)}{Wine(S)} = \frac{S^2 \cdot \frac{R6}{R3} \cdot \frac{(R4+R3)}{(R5+R6)}}{S^2 + S \cdot \frac{R5}{C1 \cdot R1 \cdot R3} \cdot \frac{(R4+R3)}{(R5+R6)} + \frac{R4}{C3 \cdot R2 \cdot C1 \cdot R1 \cdot R3}} \Rightarrow \text{High Crossings}$$

$$\frac{V2(S)}{Wine(S)} = \frac{\frac{R6}{R2 \cdot C2 \cdot C1 \cdot R1 \cdot R3} \cdot \frac{(R4+R3)}{(R5+R6)}}{S^2 + S \cdot \frac{R5}{C1 \cdot R1 \cdot R3} \cdot \frac{(R4+R3)}{(R5+R6)} + \frac{R4}{R2 \cdot C2 \cdot C1 \cdot R1 \cdot R3}} \Rightarrow \text{Low Passes}$$

$$\frac{V1(S)}{Wine(S)} = - \frac{S \cdot \frac{R6}{C1 \cdot R1 \cdot R3} \cdot \frac{(R4+R3)}{(R5+R6)}}{S^2 + S \cdot \frac{R5}{C1 \cdot R1 \cdot R3} \cdot \frac{(R4+R3)}{(R5+R6)} + \frac{R4}{C2 \cdot R2 \cdot C1 \cdot R1 \cdot R3}} \Rightarrow \text{Pass Band}$$

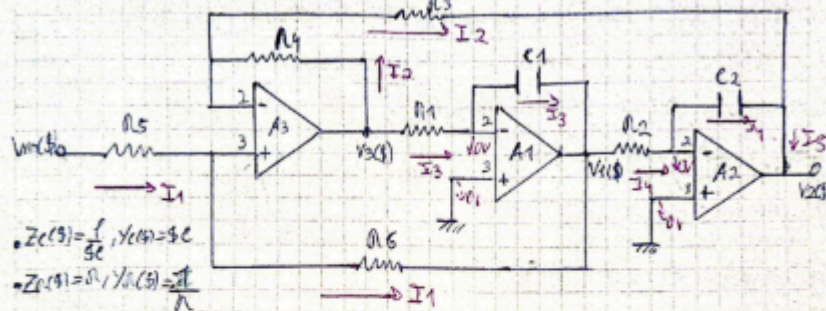
Simulations:



Simulation of the three requested transfers.

• Filtro de variables de estado, busquemos los transformadores

a) $\frac{V_2(s)}{V_{in}(s)}$, b) $\frac{V_3(s)}{V_{in}(s)}$, c) $\frac{V_1(s)}{V_{in}(s)}$



• $Z_c(s) = \frac{1}{sC}$
• $Z_r(s) = R, Y_c(s) = \frac{1}{R}$

• Suponga amplificación ideal $\Rightarrow$ Para A_3, A_1 y $A_2 \Rightarrow V^+ = V^-$

$V_{in}(s) - V_{A3}^+ V_{A3}^- I_1(s) = (V_{A3} - V_1(s)) Y_{06}(s)$

$\Rightarrow V_{in}(s) \cdot Y_{05}(s) - V_{A3}^+ Y_{05}(s) = V_{A3}^+ Y_{06}(s) - V_1(s) \cdot Y_{06}(s)$

$\Rightarrow V_1(s) \cdot Y_{06}(s) = V_{A3}^+ \cdot Y_{06}(s) + V_{A3}^+ \cdot Y_{05}(s) - V_{in}(s) \cdot Y_{05}(s)$

$V_1(s) = V_{A3}^+ \left(\frac{Y_{06}(s) + Y_{05}(s)}{Y_{06}(s)} \right) - V_{in}(s) \cdot \frac{Y_{05}(s)}{Y_{06}(s)}$

$V_1(s) = V_{A3}^+ \left(1 + \frac{Y_{05}(s)}{Y_{06}(s)} \right) - V_{in}(s) \cdot \frac{Y_{05}(s)}{Y_{06}(s)}$ (A)

• $V_{A3}^+ = V_{A3}^- \Rightarrow I_2(s) = (V_3(s) - V_{A3}^-) \cdot Y_{04}(s) = (V_{A3} - V_2(s)) \cdot Y_{04}(s)$

$\Rightarrow V_3(s) \cdot Y_{04}(s) - V_{A3}^+ Y_{04}(s) = V_{A3}^+ \cdot Y_{04}(s) - V_2(s) \cdot Y_{04}(s)$

$V_3(s) \cdot Y_{04}(s) + V_2(s) \cdot Y_{04}(s) = V_{A3}^+ \cdot Y_{04}(s) + V_{A3}^+ \cdot Y_{04}(s)$

$V_3(s) \cdot Y_{04}(s) + V_2(s) \cdot Y_{04}(s) = V_{A3}^+ (Y_{04}(s) + Y_{04}(s))$ (B)

• $I_3(s) = (V_3(s) - V_{A1}^-) \cdot Y_{01}(s) = (V_{A1} - V_1(s)) \cdot Y_{01}(s)$, $V_{A1}^- = V_{A1}^+ = 0V$

$\Rightarrow (V_3(s) - 0V) \cdot Y_{01}(s) = (0V - V_1(s)) \cdot Y_{01}(s)$

$\Rightarrow V_3(s) \cdot Y_{01}(s) = -V_1(s) \cdot Y_{01}(s)$

$V_3(s) = -V_1(s) \cdot \frac{Y_{01}(s)}{Y_{01}(s)}$ (C)

• $I_4(s) = (V_1(s) - V_{A2}^-) \cdot Y_{02}(s) = (V_{A2} - V_2(s)) \cdot Y_{02}(s)$, $V_{A2}^- = V_{A2}^+ = 0V$

$\Rightarrow (V_1(s) - 0V) \cdot Y_{02}(s) = (0V - V_2(s)) \cdot Y_{02}(s)$

$\Rightarrow V_1(s) \cdot Y_{02}(s) = -V_2(s) \cdot Y_{02}(s)$

$\Rightarrow V_1(s) \cdot \frac{Y_{02}(s)}{Y_{02}(s)} = -V_2(s)$ (D)

• Reemplazamos (C) y (D) en (B)

$-V_1(s) \cdot \frac{Y_{01}(s)}{Y_{01}(s)} \cdot Y_{04}(s) - V_1(s) \cdot \frac{Y_{02}(s)}{Y_{02}(s)} \cdot Y_{04}(s) = V_{A3}^+ (Y_{04}(s) + Y_{04}(s))$

$$-V_1(s) \cdot \frac{Y_1(s) \cdot Y_4(s)}{Y_1(s)} - V_1(s) \frac{Y_2(s) \cdot Y_3(s)}{Y_2(s)} = V_{A3}^+$$

$$\frac{-V_1(s) \cdot Y_4(s)}{Y_3(s) + Y_4(s)} - V_1(s) \frac{Y_3(s)}{Y_3(s) + Y_4(s)} = V_{A3}^+$$

$$\boxed{-V_1(s) \frac{Y_1(s) \cdot Y_4(s)}{Y_1(s)(Y_3(s) + Y_4(s))} - V_1(s) \frac{Y_2(s) \cdot Y_3(s)}{Y_2(s)(Y_3(s) + Y_4(s))} = V_{A3}^+} \quad (E)$$

Simplifico (E) en (A)

$$V_1(s) = \left[-V_1(s) \frac{Y_1(s) \cdot Y_4(s)}{Y_1(s)(Y_3(s) + Y_4(s))} - V_1(s) \frac{Y_2(s) \cdot Y_3(s)}{Y_2(s)(Y_3(s) + Y_4(s))} \right] \cdot \left(\frac{1 + Y_5(s)}{Y_6(s)} \right) - \frac{V_{m1}(s) \cdot Y_5(s)}{Y_6(s)}$$

• Por (E) $\Rightarrow a = \frac{Y_1(s) \cdot Y_4(s)}{Y_1(s)(Y_3(s) + Y_4(s))}$

• Por (E) $\Rightarrow b = \frac{Y_2(s) \cdot Y_3(s)}{Y_2(s)(Y_3(s) + Y_4(s))}$

• Por (E) $\Rightarrow c = 1 + \frac{Y_5(s)}{Y_6(s)}$, • Por (E) $\Rightarrow d = \frac{Y_5(s)}{Y_6(s)}$

$$\Rightarrow V_1(s) = -V_1(s) \cdot a \cdot c - V_1(s) \cdot b \cdot c - V_{m1}(s) \cdot d$$

$$\Rightarrow V_1(s) + V_1(s) \cdot a \cdot c + V_1(s) \cdot b \cdot c = -V_{m1}(s) \cdot d$$

$$\Rightarrow V_1(s) [1 + a \cdot c + b \cdot c] = -V_{m1}(s) \cdot d$$

$$\Rightarrow \boxed{\frac{V_1(s)}{V_{m1}(s)} = \frac{-d}{1 + a \cdot c + b \cdot c}} \quad (F)$$

• Además $\Rightarrow \boxed{V_2(s) = \frac{V_{m1}(s) \cdot d}{1 + a \cdot c + b \cdot c} \cdot \frac{Y_2(s)}{Y_2(s)}} \quad (G)$

• Además $\Rightarrow \boxed{V_3(s) = \frac{V_{m1}(s) \cdot d}{1 + a \cdot c + b \cdot c} \cdot \frac{Y_3(s)}{Y_3(s)}} \quad (H)$

• Para trabajar (F), (G) y (H) hay que tener los valores

La expresión: $\frac{-d}{1 + a \cdot c + b \cdot c}$

$$a = \frac{Y_1(s) \cdot Y_4(s)}{Y_1(s)(Y_3(s) + Y_4(s))} = \frac{g_{C1} \cdot 1/n_4}{\frac{1}{n_1} \cdot \left(\frac{1}{n_3} + \frac{1}{n_4} \right)} \cdot \frac{n_4}{n_4} = \frac{g_{C1}}{\frac{1}{n_1} \cdot \left(\frac{1}{n_3} + \frac{1}{n_4} \right)}$$

$$a = \frac{g_{C1}}{\frac{1}{n_1} + \frac{1}{n_1}} = \frac{g_{C1}}{\frac{n_4 + n_3}{n_3 \cdot n_4}} = \boxed{\frac{g_{C1} \cdot n_3 \cdot n_4}{n_4 + n_3}} \quad a$$

$$b = \frac{Y_2(s) \cdot Y_3(s)}{Y_2(s)(Y_3(s) + Y_4(s))} = \frac{\frac{1}{n_2} \cdot \frac{1}{n_3}}{g_{C2} \left(\frac{1}{n_3} + \frac{1}{n_4} \right)} = \frac{1}{g_{C2} \cdot n_3 \cdot n_4 \left(\frac{1}{n_3} + \frac{1}{n_4} \right)} \quad b$$

$$b = \frac{1}{g_{C2} \cdot n_3 \cdot n_4 \left(\frac{n_4 + n_3}{n_3 \cdot n_4} \right)} = \frac{1}{g_{C2} \cdot n_2 \cdot (n_4 + n_3)} = \boxed{\frac{n_4}{g_{C2} \cdot n_2 \cdot (n_4 + n_3)}} \quad b$$

$$c = 1 + \frac{r_{AS}(t)}{r_{AB}(t)} = 1 + \frac{r_{AS}}{r_{AB}} = \boxed{1 + \frac{r_6}{r_5}} \quad (c)$$

$$d = \frac{r_{AS}(t)}{r_{AB}(t)} = \frac{r_{AS}}{r_{AB}} = \boxed{\frac{r_6}{r_5}} \quad (d)$$

• Ahorro Inverso:

$$a \cdot c = \frac{\$ C_1 \cdot r_1 \cdot r_3}{r_4 + r_3} \cdot \left(1 + \frac{r_6}{r_5}\right) = \frac{\$ C_1 \cdot r_1 \cdot r_3}{(r_4 + r_3)} \cdot \frac{(r_5 + r_6)}{r_5} = \frac{\$ C_1 \cdot r_1 \cdot r_3}{r_5} \cdot \frac{(r_5 + r_6)}{(r_4 + r_3)}$$

$$b \cdot c = \frac{r_4}{\$ C_2 \cdot r_2 \cdot (r_4 + r_3)} \cdot \left(1 + \frac{r_6}{r_5}\right) = \frac{r_4}{\$ C_2 \cdot r_2 \cdot (r_4 + r_3)} \cdot \frac{(r_5 + r_6)}{r_5} = \frac{r_4}{\$ C_2 \cdot r_2 \cdot r_5} \cdot \frac{(r_5 + r_6)}{(r_4 + r_3)}$$

• Retomando \$

$$\frac{V_1(t)}{V_{inv}(t)} = - \frac{\$ \frac{r_6}{r_5}}{1 + \frac{\$ C_1 \cdot r_1 \cdot r_3}{r_5} \cdot \frac{(r_5 + r_6)}{(r_4 + r_3)} + \frac{r_4}{\$ C_2 \cdot r_2 \cdot r_5} \cdot \frac{(r_5 + r_6)}{(r_4 + r_3)}} \cdot \frac{\$}{\$}$$

$$\frac{V_1(t)}{V_{inv}(t)} = - \frac{[\$ \frac{r_6}{r_5}]}{\left[\$^2 \frac{C_1 \cdot r_1 \cdot r_3}{r_5} \cdot \frac{(r_5 + r_6)}{(r_4 + r_3)} + \$ + \frac{r_4}{C_2 \cdot r_2 \cdot r_5} \cdot \frac{(r_5 + r_6)}{(r_4 + r_3)} \right]} \cdot \frac{(\$ \frac{r_5(r_4 + r_3)}{C_1 \cdot r_1 \cdot r_3 \cdot r_5})}{(\$ \frac{r_5(r_4 + r_3)}{C_1 \cdot r_1 \cdot r_3 \cdot r_5})}$$

$$\frac{V_1(t)}{V_{inv}(t)} = - \frac{\$ \frac{r_6(r_4 + r_3)}{C_1 \cdot r_1 \cdot r_3 \cdot (r_5 + r_6)}}{\$^2 + \$ \frac{r_5 \cdot (r_4 + r_3)}{C_1 \cdot r_1 \cdot r_3 \cdot (r_5 + r_6)} + \frac{r_4}{C_2 \cdot r_2 \cdot C_1 \cdot r_1 \cdot r_3}} \rightarrow \text{Nto c)}$$

• Corroborando $\frac{V_1(t)}{V_{inv}(t)}$ solo resta desarrollar la expresión
excepto el signo "-" para luego multiplicar por $\frac{r_4 \cdot r_5}{r_2 \cdot r_3}$

luego $\frac{V_2(t)}{V_{inv}(t)}$, ya multiplicamos por $\frac{V_1(t)}{V_{inv}(t)}$ para obtener $\frac{V_2(t)}{V_{inv}(t)}$.

$$\frac{V_2(t)}{V_{inv}(t)} = \frac{\$ \frac{r_6(r_4 + r_3)}{C_1 \cdot r_1 \cdot r_3 \cdot (r_5 + r_6)}}{\$^2 + \$ \frac{r_5 \cdot (r_4 + r_3)}{C_1 \cdot r_1 \cdot r_3 \cdot (r_5 + r_6)} + \frac{r_4}{C_2 \cdot r_2 \cdot C_1 \cdot r_1 \cdot r_3}} \cdot \frac{1/r_2}{\$ \cdot r_2}$$

$$\frac{V_2(t)}{V_{inv}(t)} = \frac{\$ \cdot \frac{r_6(r_4 + r_3)}{r_2 \cdot C_1 \cdot r_1 \cdot r_3 \cdot (r_5 + r_6)}}{\$^3 + \$^2 \cdot \frac{C_2 \cdot r_5}{C_1 \cdot r_1 \cdot r_3} \cdot \frac{(r_4 + r_3)}{(r_5 + r_6)} + \$ \cdot \frac{r_4}{r_2 \cdot C_1 \cdot r_1 \cdot r_3}} \cdot \frac{1/r_2}{1/r_2}$$

$$\frac{V_2(t)}{V_{inv}(t)} = \frac{\$ \frac{r_6}{r_2 \cdot C_1 \cdot r_1 \cdot r_3} \cdot \frac{(r_4 + r_3)}{(r_5 + r_6)}}{\$^3 + \$^2 \cdot \frac{r_5}{C_1 \cdot r_1 \cdot r_3} \cdot \frac{(r_4 + r_3)}{(r_5 + r_6)} + \$ \frac{r_4}{r_2 \cdot C_1 \cdot r_1 \cdot r_3}}$$

(Nto b)

$$\bullet \frac{V_3(\$)}{V_{im}(\$)} = \frac{\$ \frac{n_6(n_4+n_3)}{c_1 n_1 n_3 (n_5+n_6)}}{\$^2 + \$ \frac{n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \frac{n_4}{c_2 n_2 \cdot c_1 n_1 n_3}} \cdot \frac{\$ c_1}{1/n_1}$$

$$\bullet \frac{V_3(\$)}{V_{im}(\$)} = \frac{\$ \frac{n_6}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} \cdot \$ c_1 n_1}{\$^2 + \$ \frac{n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \frac{n_4}{c_2 n_2 \cdot c_1 n_1 n_3}}$$

$$\bullet \frac{V_3(\$)}{V_{im}(\$)} = \frac{\$^2 \cdot \frac{n_6}{n_3} \frac{(n_4+n_3)}{(n_5+n_6)}}{\$^2 + \$ \frac{n_5}{c_1 n_1 n_3} \frac{(n_4+n_3)}{(n_5+n_6)} + \frac{n_4}{c_2 n_2 \cdot c_1 n_1 n_3}}$$

balas)

Exercise #6

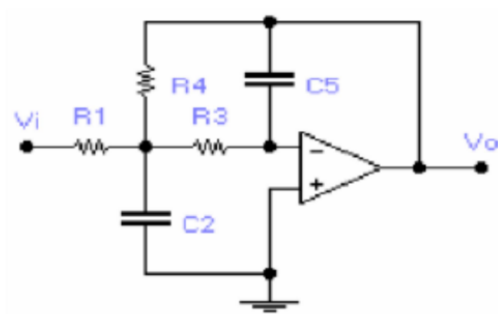


Imagen 20

Circuit for analysis.

a) Determine the voltage transfer of the filter, which is specified in the circuit.

b) Recalculate the value of the components that make up the circuit if $\omega_0 = 1000 \text{ rad/s}$ is desired and capacitors of 4700pF and 47pF are available.

$$a) \quad \frac{V_o}{V_i} = - \frac{1}{R_1 R_3 C_2 C_5} \frac{1}{s^2 + s \left(\frac{C_5 R_1 R_4 + C_5 R_3 R_1 + C_5 R_3 R_4}{R_1 R_3 R_4 C_2 C_5} \right) + \frac{1}{R_3 R_4 C_2 C_5}} = - \frac{100}{s^2 + s \cdot 3 + 100}$$

b)

$$R_1 = 2.1276 \text{ M}\Omega \quad C_2 = 4700 \text{ pF} \quad R_3 = 2.1276 \text{ M}\Omega \quad R_4 = 2.1276 \text{ M}\Omega \\ C_5 = 47 \text{ pF}$$

Ejercicio 6

a) Planteo de ecuaciones iniciales:

- $I_1 = I_2 + I_3 + I_4$ (1)
- $I_1 = \frac{V_{in} - V_x}{R_1}$ (2)
- $I_2 = \frac{V_x - 0V}{R_3} = \frac{0V - V_o}{Z_{C5}}$ Como R_3 y C_5 están en serie, entonces $I_2 = \frac{V_x - V_o}{R_3 + Z_{C5}}$ (3)
- Después de V_x de la siguiente igualdad:
- $\frac{V_x - 0V}{R_3} = \frac{0V - V_o}{Z_{C5}} \rightarrow \frac{V_x}{R_3} = - \frac{V_o}{Z_{C5}} \rightarrow V_x = - \frac{V_o \cdot R_3}{Z_{C5}}$ (4)
- $I_3 = \frac{V_x - V_o}{R_4}$ (5)
- $I_4 = \frac{V_x - 0V}{Z_{C2}} = \frac{V_x}{Z_{C2}}$ (6)
- En la ecuación (1) reemplazo a las corrientes por (2), (3), (5) y (6).

$$\frac{V_{in} - V_x}{R_1} = \frac{V_x - V_0}{R_3 + \frac{1}{sC_5}} + \frac{V_x - V_0}{R_4} + \frac{V_x}{\frac{1}{sC_2}}$$

Reemplazando a V_x por (4):

$$\frac{V_{in}}{R_1} - \frac{(-V_0 \cdot sC_5 \cdot R_3)}{R_1} = \frac{(-V_0 \cdot sC_5 \cdot R_3 - V_0)}{\frac{sC_5 R_3 + 1}{sC_5}} + \frac{(-V_0 \cdot sC_5 \cdot R_3 - V_0)}{R_4} + \frac{(-V_0 \cdot sC_5 \cdot R_3 - V_0)}{\frac{1}{sC_2}}$$

$$\frac{V_{in}}{R_1} + \frac{V_0 \cdot sC_5 \cdot R_3}{R_1} = -V_0 \cdot \frac{(sC_5 R_3 + 1) \cdot sC_5}{(sC_5 R_3 + 1)} - \frac{V_0 (sC_5 R_3 + 1)}{R_4} - \frac{V_0 \cdot sC_5 \cdot R_3 \cdot sC_2}{R_4}$$

$$\frac{V_{in}}{R_1} = -V_0 \cdot \left(sC_5 + \frac{sC_5 R_3 + 1}{R_4} + \frac{sC_5 R_3 \cdot sC_2}{R_4} + \frac{sC_5 R_3}{R_4} \right)$$

Entonces el $H(s)$ sería:

$$H(s) = \frac{V_0(s)}{V_{in}(s)} = - \frac{1}{R_1 \left(sC_5 + \frac{sC_5 R_3 + 1}{R_4} + \frac{sC_5 R_3 \cdot sC_2}{R_4} + \frac{sC_5 R_3}{R_4} \right)}$$

$$H(s) = - \frac{1}{sC_5 R_1 + \frac{sC_5 R_3 R_1}{R_4} + \frac{R_1}{R_4} + \frac{sC_5 R_3 \cdot sC_2 R_1}{R_4} + \frac{sC_5 R_3 R_1}{R_4}}$$

$$H(s) = - \frac{1}{\left[s^2 C_5 R_3 C_2 R_1 + s \left(C_5 R_1 + \frac{C_5 R_3 R_1}{R_4} + \frac{C_5 R_3 R_1}{R_4} \right) + \frac{R_1}{R_4} \right]} \cdot \frac{C_5 R_3 C_2 R_1}{\frac{1}{C_5 R_3 C_2 R_1}}$$

$$H(s) = - \frac{1}{C_5 R_3 C_2 R_1} \cdot \frac{1}{s^2 + s \left(C_5 R_1 + \frac{C_5 R_3 R_1}{R_4} + \frac{C_5 R_3 R_1}{R_4} \right) + \frac{1}{C_5 R_3 C_2 R_1} + \frac{R_1}{R_4} \cdot \frac{1}{C_5 R_3 C_2 R_1}}$$

$$H(s) = - \frac{1}{C_5 R_3 C_2 R_1} \cdot \frac{1}{s^2 + s \left(\frac{C_5 R_1 R_4 + C_5 R_3 R_1 + C_5 R_3 R_1}{R_4} \right) + \frac{1}{R_4 C_5 R_3 C_2}}$$

Reemplazando los datos: $R_1 = R_3 = R_4 = 1$; $C_2 = 1$; $C_5 = 0.01$

$$H(s) = - \frac{1}{0.01} \cdot \frac{1}{s^2 + s \left(\frac{0.01 + 0.01 + 0.01}{0.01} \right) + \frac{1}{0.01}}$$

$$H(s) = - \frac{100}{s^2 + s \cdot 3 + 100}; \text{ Parámetros de 2º orden}$$

b) Dado que los valores que nos brindaron de las impedancias para el punto a) se encuentran normalizados en impedancia y en frecuencia (caso de los capacitores). Hoy que obtenemos la norma de normalización de impedancia y frecuencia para así poder recalcular el circuito para los valores solicitados.

Como la ω_0 requerida es: $\omega_0 = 1000 \text{ rad/s}$ (ω_0 desnormalizado)

Y sabíamos del punto a) que: $\omega_n \text{ norma} = 10$

Entonces se obtiene que N_z vale: $\frac{\omega_{desn}}{\omega_n \text{ norma}} = 100 = N_z$

Luego, recordando que el valor normalizado en frecuencia e impedancia del capacitor se calcula como:

$$C_N = C \cdot N_z \cdot N_z \rightarrow \text{despejando } N_z: N_z = \frac{C_N}{C \cdot N_z}$$

$$\text{y reemplazando valores: } N_z = \frac{1F}{4700pF \cdot 100} = 2,1276 \times 10^6$$

Finalmente recalculamos los valores desnormalizados de R_1, R_2 y R_3 para una $\omega_0 = 1000 \text{ rad/s}$.

$$R_{1N} = 1\Omega \rightarrow R_1 = R_{1N} \cdot N_z = 2,1276 \times 10^6 \Omega$$

$$R_{2N} = 1\Omega \rightarrow R_2 = R_{2N} \cdot N_z = 2,1276 \times 10^6 \Omega$$

$$R_{3N} = 1\Omega \rightarrow R_3 = R_{3N} \cdot N_z = 2,1276 \times 10^6 \Omega$$

Simulation

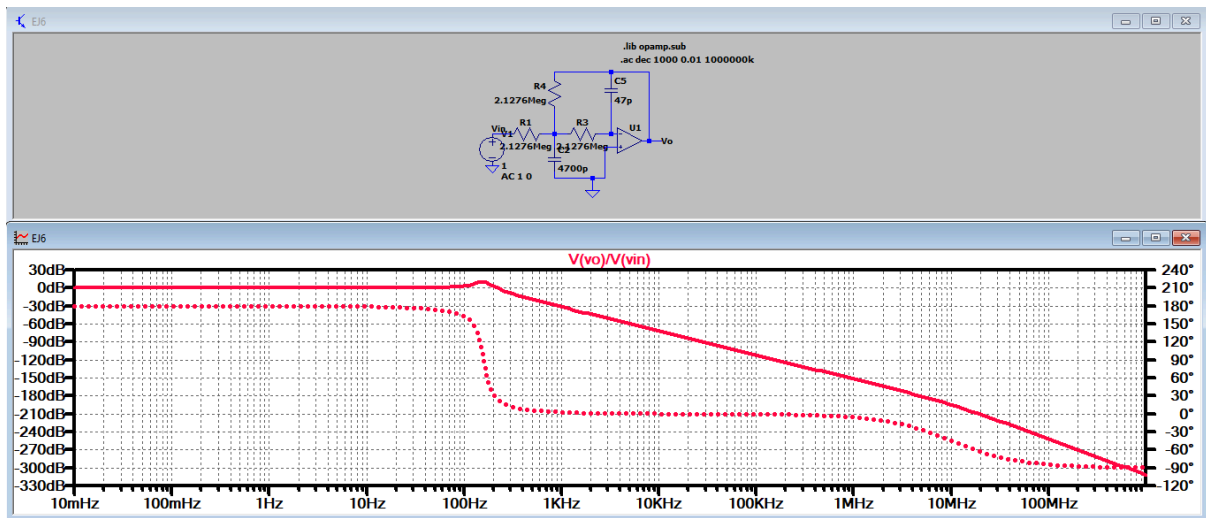


Imagen 21

Voltage transfer simulation.

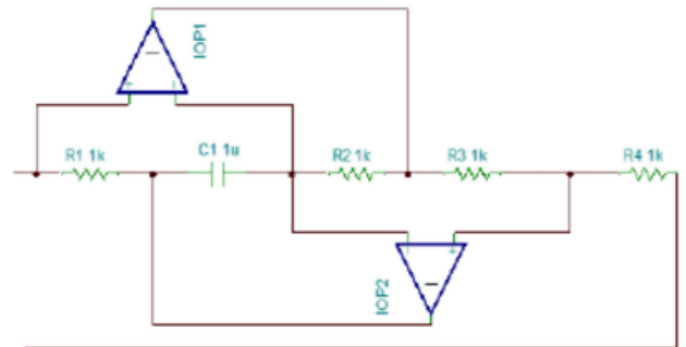
Exercise #7

Determine the excitation impedance. Plot the magnitude and phase of the impedance, interpreting the results.

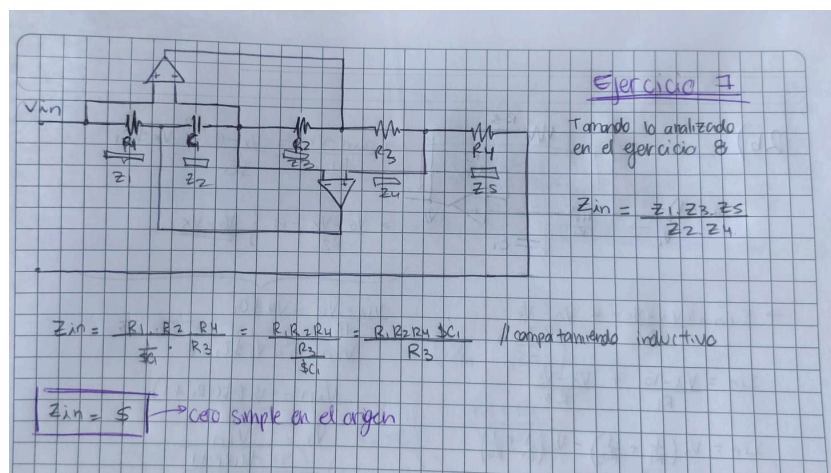
$$S_{entree} = \frac{C_{ome}}{I_{in}} = \frac{Z_1 \cdot Z_3 \cdot Z_5}{Z_2 \cdot Z_4} = \frac{R_1 \cdot R_2 \cdot R_4}{R_3 \cdot \frac{1}{s \cdot C_1}}$$

$$= \frac{s \cdot R_1 \cdot R_2 \cdot R_4 \cdot C_1}{R_3} = S$$

Imagen 22



Active dipole for analysis.



Simulation

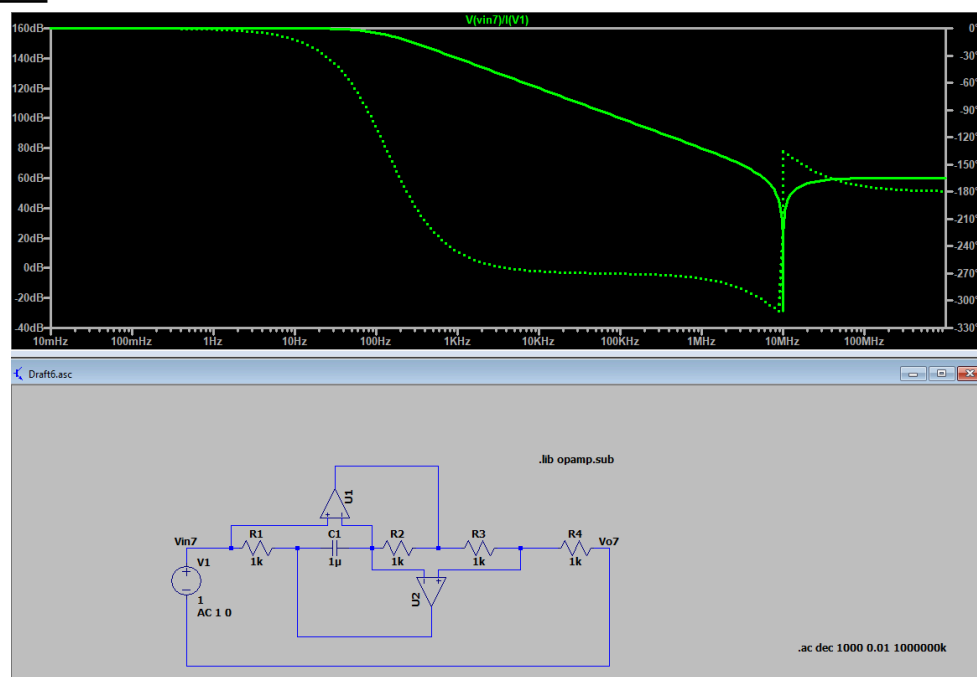


Imagen 23

Input impedance simulation.

Exercise #8

For the following active dipole, determine the excitation impedance. Plot the magnitude and phase of the impedance, interpreting the results.

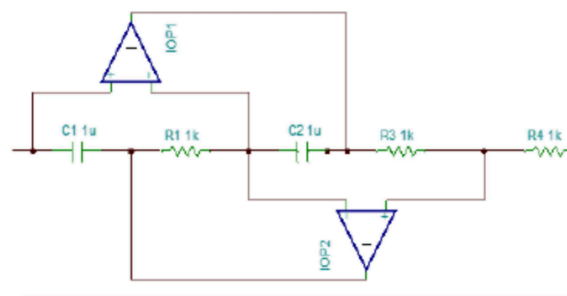


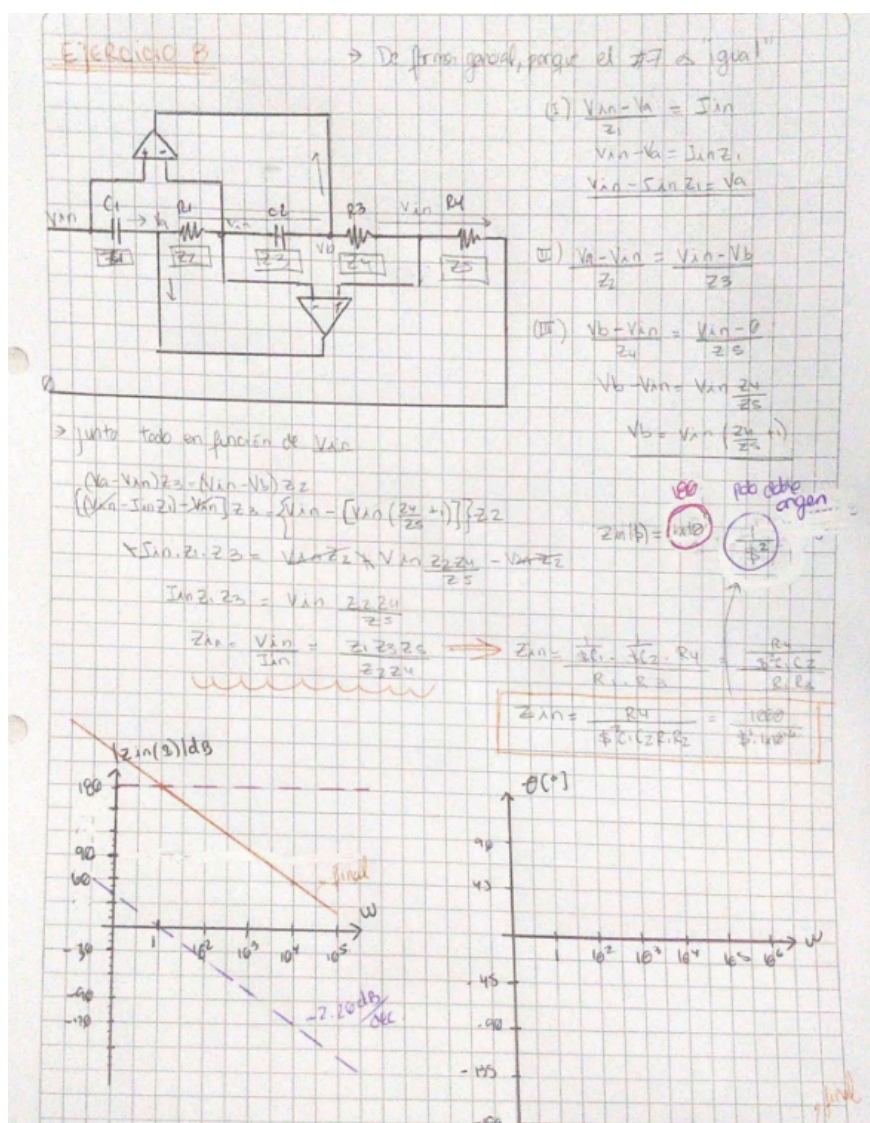
Imagen 24

$$\text{Sentence} = \frac{Come}{I_{in}} = \frac{Z1 \cdot Z3 \cdot Z5}{Z2 \cdot Z4} = \frac{R4 \cdot \frac{1}{s \cdot C2} \cdot \frac{1}{s \cdot C1}}{R1 \cdot R3}$$

$$= \frac{R4}{s^2 \cdot C1 \cdot C2 \cdot R1 \cdot R3} = \frac{1}{s^2 \cdot 1 \times 10^{-9}}$$

Active dipole for analysis.

According to the Bode plot as a function of frequency, it would appear to behave like a low-pass filter. Thanks to this network, we can construct an FDNR as we saw in class.



Simulation

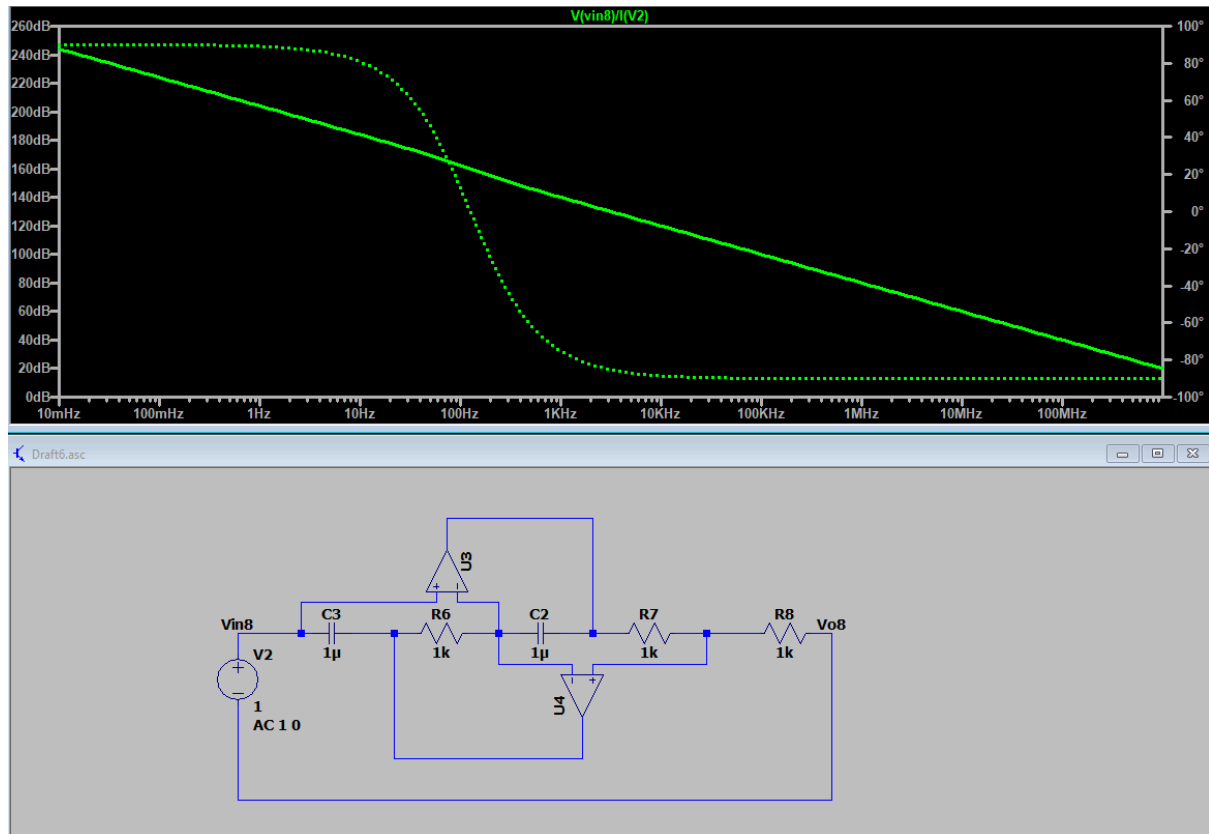


Imagen 25

Input impedance simulation.

Exercise #9

For the following active dipole, determine the excitation impedance. Indicate/propose applications for this network.

$$S_{\text{entree}} = -\frac{R_1 \cdot Z_L}{R_2}$$

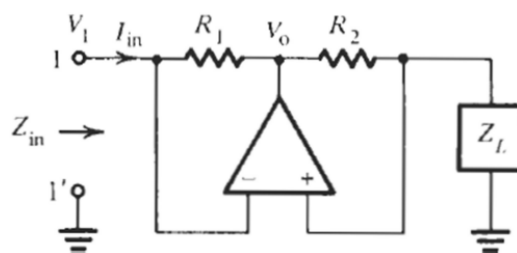
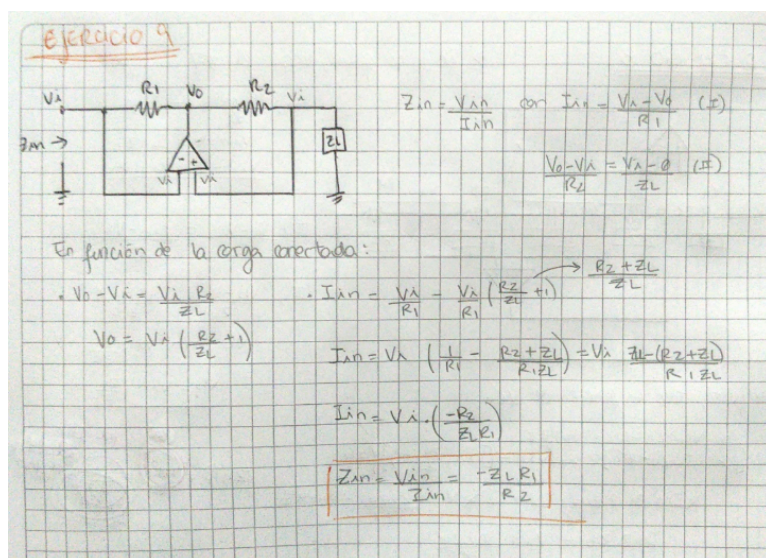


Imagen 26

Active dipole for analysis



Simulation:

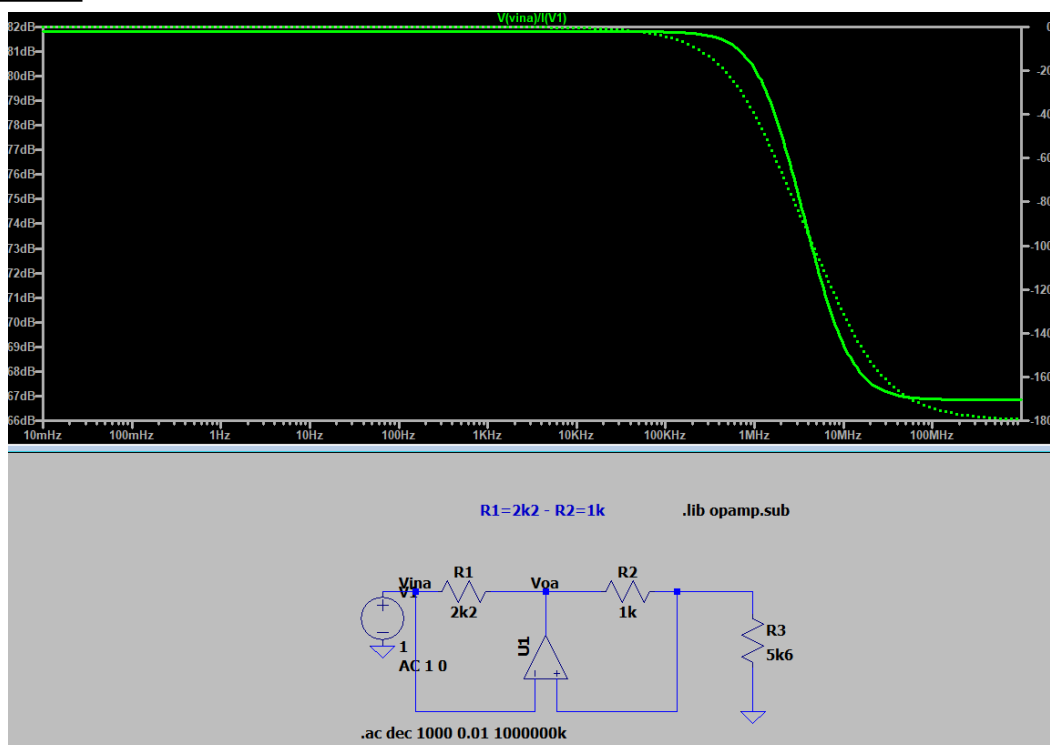


Imagen 27

Simulation of the excitation impedance for a purely resistive load.

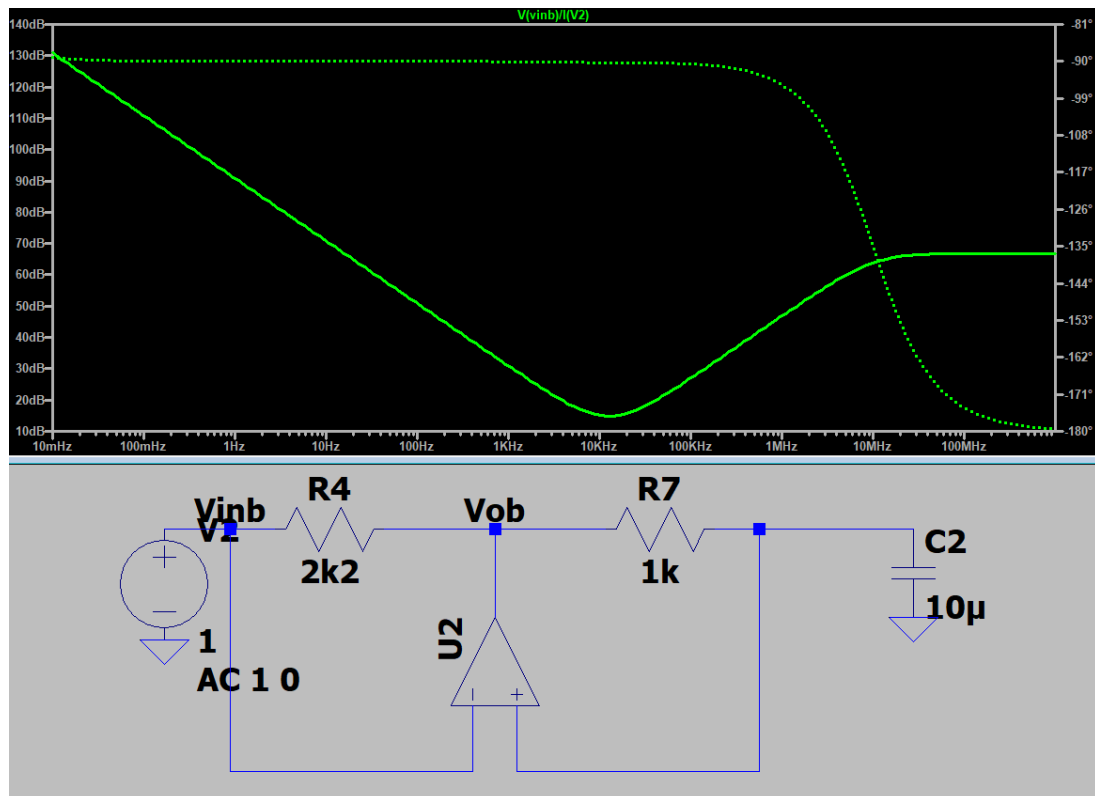


Imagen 28

Simulation of the excitation impedance for a purely capacitive load.

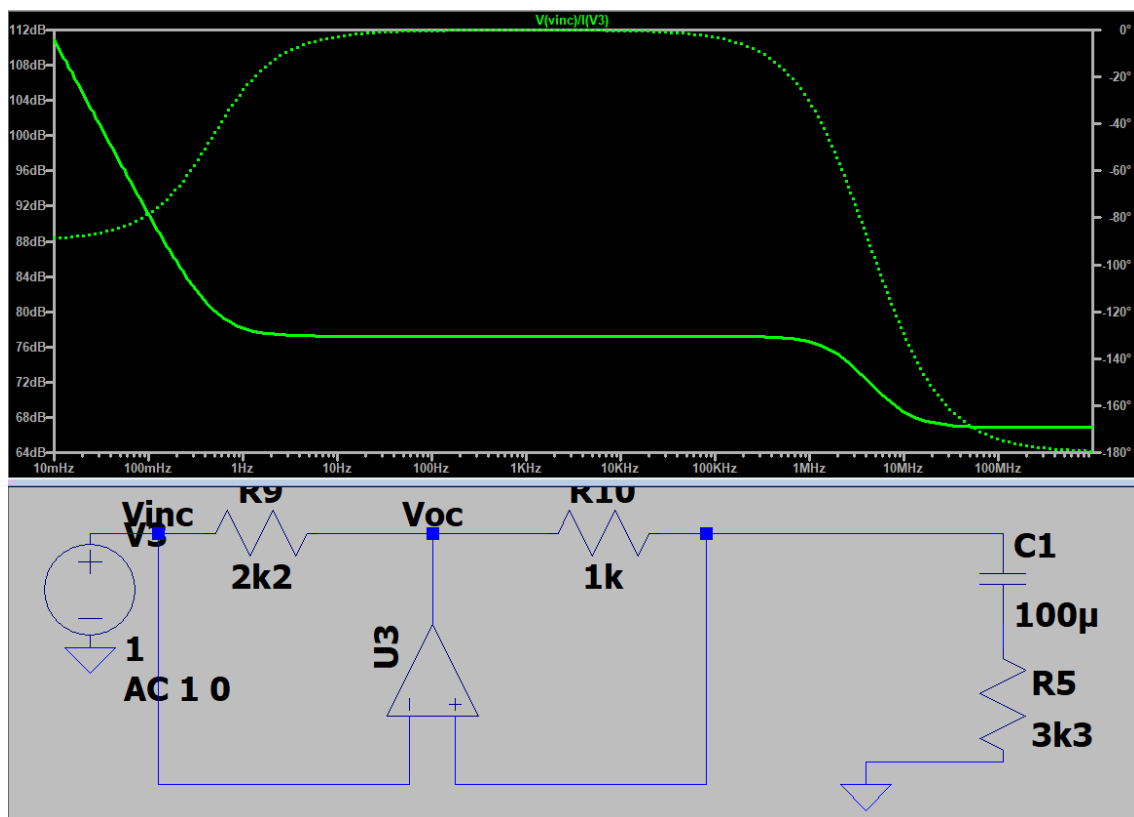


Imagen 29

Simulation of excitation impedance for a capacitive and resistive load.

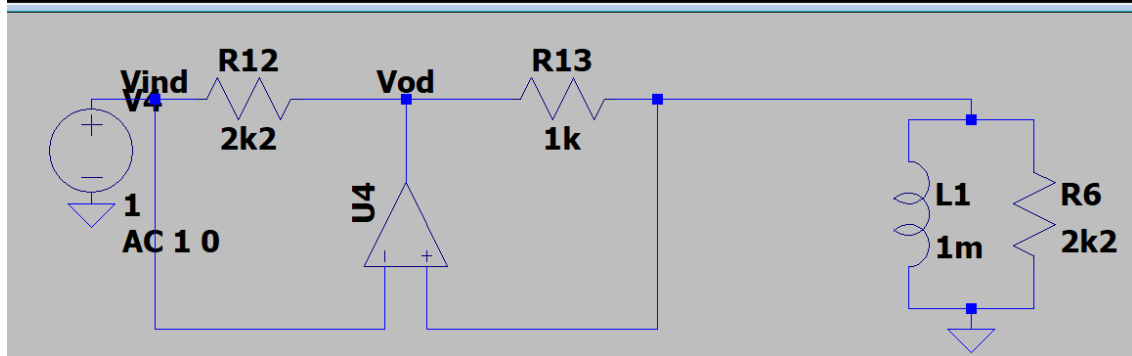
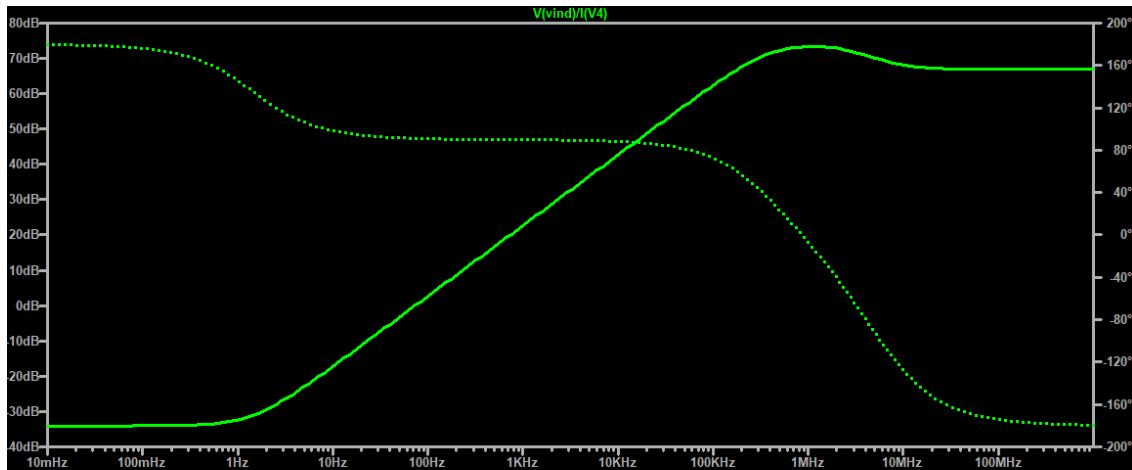


Imagen 30

Simulation of the excitation impedance for an impedance and resistive load.